

NOW YOU SEE IT, NOW YOU DON'T: RETHINKING VISUAL TOKEN ALLOCATION ACROSS VISION LANGUAGE MODEL LAYERS

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ABSTRACT

Vision-language models (VLMs) can often localize relevant regions in an image, yet still fail to perceive the fine-grained visual evidence needed to answer correctly, leading to visual hallucinations. Using more tokens provides a finer representation of the image, but also increases the computation, memory, and latency required for inference. VLMs therefore operate under a limited visual-token budget: spending more of it on irrelevant details wastes computation without improving the answer. The central challenge is therefore to spend this limited budget where additional visual detail is most useful.

A common strategy is attention-guided cropping, however, existing approaches typically extract this attention from a manually selected layer or a fixed range of layers, without establishing which depth provides the most reliable signal for visual selection. We show that the choice of layer is a dominant factor in attention-guided visual selection. By systematically masking text-to-image attention across network depth, we show that the information relevant to the model’s prediction is concentrated within an early-to-middle portion of the network. This boundary occurs substantially earlier than the layers used by existing attention-based selection methods. Based on this finding, we introduce two complementary strategies. Depth-Weighted Attention (DWA) combines attention information across the full depth of the model through a closed-form ridge regression, learning layer weights that suppress less informative late-layer contributions and produce more reliable spatial selection. Adaptive Visual Resolution (AVR) uses the identified depth boundary to remove visual tokens that no longer affect the prediction and redirects the saved computation toward higher-resolution encoding, allowing the model to represent important image regions with finer visual detail.

Across nine matched-compute comparisons, DWA consistently improves over existing layer-selection strategies, achieving gains of 7.9 and 8.9 percentage points over the strongest baseline and locating answer-relevant evidence in 63.9% of cases compared with 13.6%. Against native dynamic-resolution models operating at $5.5\times$ the visual-token budget, DWA and AVR achieve statistically indistinguishable performance while using only 14–18% of the visual-token computation.

1 INTRODUCTION

A vision–language model asked “what material is the glove?” about a 2250×1500 photograph must first encode the image into visual tokens. At any finite budget the glove may occupy a fraction of a token, and the evidence is gone before the language model runs. The model still *localises* it, its attention concentrates on the right region but it cannot *perceive* it.

The standard remedy is attention-guided cropping: run the model once, read its attention over image positions, crop where the attention peaks, and re-encode that crop at higher density (Zhang et al., 2023; Shen et al., 2024). Every method in this family faces one decision that is rarely examined: **at which layer should the attention map be read?** Published methods pick a single mid-network layer or average a fixed block of late layers.

We identify the depth at which image information reaches the answer and find that attention-guided cropping methods extract their signal entirely after this point. Masking text tokens’ attention to image tokens layer by layer shows that image-to-text transport occurs within the first two thirds of the network. The answer-emission token does not directly attend to the image; instead, it receives information through text tokens that have aggregated visual evidence earlier in the network. Blocking all image access after layer 16 of 28 does not change any answers. The same transition occurs at a normalized depth of 0.57 in both the 28-layer and 36-layer models.

Three findings follow from these measurements:

- **(i) The signal used by this family is partly a residual and partly an artifact.** The conventional block-mean read-out places its crop in the last column of the token grid, immediately before a raster row-wrap, in 61.8% and 84.8% of items, compared with a 5.0% chance rate. It covers the relevant evidence in only 13.6% of items, whereas our read-out covers it in 63.9%.
- **(ii) The reported gains disappear under matched compute.** At equal visual-token budgets, published placement rules provide gains ranging from -1.6 to $+3.1$ points relative to using the same budget for higher resolution. They also remain $9.9 - 14.2$ points below the same model evaluated at its native dynamic resolution.
- **(iii) We introduce two allocation policies based on these measurements.** Depth-Weighted Attention (DWA) combines attention information across the full depth of the model through a closed-form ridge regression, learning layer weights that suppress less informative late-layer contributions and produce more reliable spatial selection. Adaptive Visual Resolution (AVR) removes 90% of visual tokens at the measured boundary, where the identity of the retained tokens has little effect, and reallocates the saved compute to higher-resolution encoding. Both reach native dynamic-resolution performance using only 14–18% of its visual-token compute. The remaining gap is therefore primarily a placement problem rather than a compute problem: a correctly placed 600-token crop exceeds native resolution by 11.6 and 16.2 points, respectively, setting a ceiling that existing placement rules do not approach.

2 RELATED WORK

Attention-guided cropping. ViCrop (Zhang et al., 2023) uses the peak of a fixed mid-layer attention map to determine where to crop, while ZoomEye (Shen et al., 2024) searches over a tree of candidate crops through repeated model queries. VisLens (He et al., 2026) takes a different approach and does not rely on attention. Instead, it decodes visual tokens using a tuned logit lens, matches them against the query, and feeds the selected crop back together with the original image in a single pass. A common limitation of these approaches is that the read-out depth is fixed manually, and their evaluations do not control for the compute that could instead be spent on simply increasing image resolution.

Layer-adaptive read-outs. LASER (Zhu et al., 2026) selects an attention layer separately for each sample by comparing attention maps obtained with and without the query. Their results suggest that there is no single ‘magic layer,’ a finding that we reach independently and investigate further from a causal perspective. ILVAD (Xie et al., 2026) aggregates newly activated tokens across neighboring layers, while TRACE (Ye et al., 2026) identifies a three-stage pattern across depth and modifies it using trained modules. ViRGo (Tran et al., 2026) also reports a finding consistent with our cross-instance analysis: ‘patch-based zooming ... can split large objects and destroy global spatial context.’ It addresses this issue with a three-way router based on object scale. We view these works as concurrent efforts that arrive at closely related conclusions. Our approach differs in three main ways: (i) we causally localize the depth range responsible for visual information transport rather than choosing a layer based only on correlation; (ii) instead of selecting a single layer, we learn a signed *re-weighting* across the full network depth; and (iii) we evaluate under matched compute budgets and across stratified settings, including comparisons against native image resolution.

Visual-token pruning. FastV (Chen et al., 2024) prunes visual tokens at layer 2, Pyramid-Drop (Xing et al., 2024) progressively removes tokens using a fixed pruning ratio, and HiRED (Arif et al., 2025) determines the token budget before the language model processes the input. These

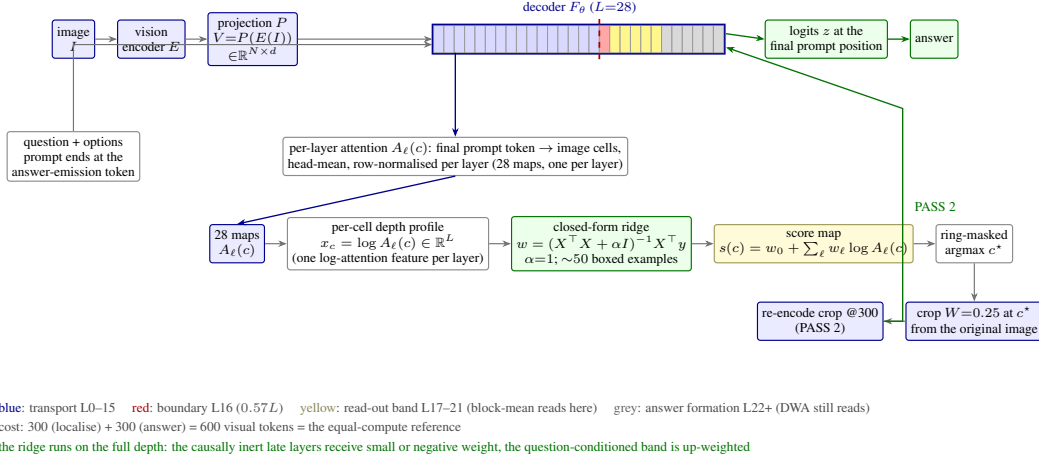


Figure 1: **VLM internals and DWA.** The image is encoded by E , projected to N visual tokens, and decoded by F_θ ; the layer bar marks the transport window (L0–15), the boundary at L16 (0.57L), the read-out band (L17–21) that block-mean methods read, and the answer-formation layers (L22+) that DWA still reads. DWA reads the final prompt token’s attention at every layer, forms each cell’s log-attention depth profile, fits a closed-form ridge to box coverage, and crops at the ring-masked arg-max of the score.

methods are primarily evaluated in terms of how much compute can be saved while retaining performance. In contrast, we derive the pruning depth directly from our measurement of visual information transport rather than treating it as a tunable hyperparameter. At the identified depth, we find that which tokens are retained matters surprisingly little: randomly selected tokens perform comparably to attention-selected tokens. Moving the pruning one layer earlier, however, produces gains of +8.4 and +8.9 points, highlighting the importance of choosing the pruning depth correctly.

3 METHODOLOGY

We first describe how a vision-language model (VLM) processes visual information and formulate the perception problem (Sec. 3.1), then introduce the framework (Sec. 3.1). We next explain depth boundary identification (Sec. 3.3), followed by Depth-Weighted Attention (DWA) for spatial selection (Sec. 3.4) and Adaptive Visual Resolution (AVR) for allocating visual computation (Sec. 3.5).

3.1 WHAT DOES A VLM SEE, AND HOW?

We begin with a frozen VLM comprising a vision encoder E , a visual-to-language mapping P , and a language decoder F_θ with L decoder layers. Here, P represents the model’s visual projection or merging components that produce embeddings for the language decoder. For a given image I and question prompt q ,

$$V = P(E(I)) \in \mathbb{R}^{N \times d}, \quad (1)$$

$$\mathbf{z} = F_\theta(V, q) \in \mathbb{R}^K. \quad (2)$$

Here, V contains N visual-token embeddings, each with dimension d . The decoder F_θ , including its output head, produces the vocabulary-logit vector $\mathbf{z}$ at the final prompt position. The vocabulary size is K , and θ denotes the pretrained model parameters, which remain frozen. The prompt q includes the question, answer options, and answer instruction supplied with the example.

The decoder’s logits define the next-token probability distribution through softmax. We use these logits to evaluate multiple-choice answers, with the scoring procedure described in Sec. 4.

3.2 VLMS AND FINE-GRAINED PERCEPTION: VISUAL RESOLUTION AS AN ENCODING BOTTLENECK

We study whether VLMS can resolve small visual details after correctly localising the relevant region, and whether allocating visual tokens to the relevant region is more effective than uniformly increasing resolution. We first sweep the target extent in merged visual tokens on single-region questions, using an oracle crop as a difficulty control (Appendix I). We find a sharp, architecture-invariant threshold at 0.15–0.25 merged tokens, corresponding to only 2–3% of the image side: accuracy jumps from 19.0% below to 78.6% above the threshold, while oracle-crop accuracy remains flat (95.8%/100.0%). Since question difficulty is unchanged across the split, the cliff reflects a failure to encode the required visual evidence rather than a gradual loss of accuracy with decreasing resolution.

We next localise the source of this failure using layer-wise decoding and representation patching (Appendices F and J). Localisation alone provides little additional decodability (72.2% vs. 68.1%), and no layer shows evidence of encoding the answer and subsequently losing it. In contrast, providing sufficient visual detail through an oracle crop corrects 94.4% of localised-but-wrong cases, despite requiring a median of only 0.14 tokens on the target. Across models, 86.7%/84.0% of errors are therefore certified as encoding failures, indicating that the dominant bottleneck is extracting fine-grained visual evidence rather than reasoning over evidence that has already been encoded.

This raises a practical question: should we increase the overall visual-token budget, or allocate the same budget more selectively? We compare oracle-targeted allocation against the best uniform resolution each model can reach, across three architectures at native resolution and the HR-Bench boundary (Appendix K). Targeted allocation is substantially more efficient: a 300-token oracle crop reaches 97.4%/93.2%, while no uniform setting up to 7,776 tokens matches it ($\geq 26 \times / 25.9 \times$ more tokens). At native resolution, a 600-token crop likewise outperforms 3,290 uniform tokens by +11.6/+16.2 points. Crucially, the benefit depends on whether the evidence is actually sub-token: targeted allocation improves V*Bench by +37.2 points, where fine details are difficult to resolve, but hurts HR-Bench 4K by −8.8 points when the evidence is already resolvable.

We evaluate two complementary strategies:

1. **Depth-Weighted Attention (DWA).** DWA uses attention information across decoder layers, together with spatial features, to score candidate crop locations through ridge regression. The selected region is cropped from the original image and re-encoded for answering.
2. **Adaptive Visual Resolution (AVR).** AVR uses the identified depth boundary to allocate visual-token computation toward higher-resolution input. It retains full visual access through the early layers and restricts access to selected visual tokens afterward.

3.3 IDENTIFYING THE DEPTH BOUNDARY

We examine where direct access to visual tokens affects answer prediction. The primary experiment uses Qwen3-VL-2B-Instruct (Qwen Team, 2025b) and Qwen2-VL-7B-Instruct (Wang et al., 2024a), each with $L = 28$ decoder layers. It processes the first 40 eligible V*Bench examples in dataset order, skipping examples without an available, nonempty bounding-box annotation. Each image is resized toward 300 visual tokens. The image, prompt, and model parameters remain unchanged between its baseline and masked runs.

We index decoder layers from zero, $\ell = 0, \dots, L-1$. Let $\mathcal{S}$ be the set of layers selected for masking. At these layers, we modify the existing additive attention mask:

$$\widetilde{M}_{u,v}^{(\ell)} = \begin{cases} m_{\min}, & \ell \in \mathcal{S}, u \in \mathcal{T}_{\text{post}}, v \in \mathcal{V}, \\ M_{u,v}^{(\ell)}, & \text{otherwise.} \end{cases} \quad (3)$$

Here, $M^{(\ell)}$ and $\widetilde{M}^{(\ell)}$ are the original and modified masks. The indices u and v denote query and key positions. The set $\mathcal{V}$ is the image-token span, identified from the input token IDs, and $\mathcal{T}_{\text{post}}$ contains all prompt positions after that span. The value $m_{\min}$ is the smallest finite value representable by the mask’s data type. This replacement is applied across attention heads and suppresses the selected connections after softmax.

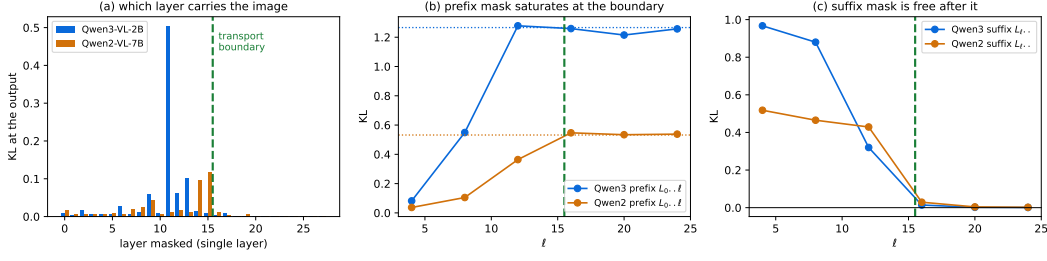


Figure 2: **Output changes under depth masking.** Mean Kullback–Leibler (KL) divergence from the unmodified next-token distribution under (a) single-layer, (b) prefix, and (c) suffix masking. Layers are zero-indexed; prefix and suffix masks both include the candidate layer. Dotted lines in panel (b) show the all-layer-mask reference for each model. The curves summarise output sensitivity across decoder depth.

We evaluate three masking schedules:

$$\mathcal{S}_{\text{single}}(b) = \{b\}, \quad (4)$$

$$\mathcal{S}_{\text{prefix}}(b) = \{0, \dots, b\}, \quad (5)$$

$$\mathcal{S}_{\text{suffix}}(b) = \{b, \dots, L - 1\}. \quad (6)$$

Here, b is the candidate layer index. Single-layer masking is evaluated at every layer. Prefix and suffix masks are evaluated at $b \in \{0, 4, 8, 12, 16, 20, 24\}$. A separate control masks all layers together. The suffix mask includes layer b itself.

We measure the effect on the next-token distribution using Kullback–Leibler (KL) divergence (Kullback & Leibler, 1951):

$$\mathbf{p} = \text{softmax}(\mathbf{z}), \quad \tilde{\mathbf{p}}_{\mathcal{S}} = \text{softmax}(\tilde{\mathbf{z}}_{\mathcal{S}}), \quad (7)$$

$$D_{\mathcal{S}} = \sum_{k=1}^K p_k \log \frac{p_k}{\tilde{p}_{\mathcal{S},k}}. \quad (8)$$

Here, $\tilde{\mathbf{z}}_{\mathcal{S}}$ is the masked logit vector. The vectors $\mathbf{p}$ and $\tilde{\mathbf{p}}_{\mathcal{S}}$ contain baseline and masked probabilities over the vocabulary, and k indexes vocabulary tokens. The scalar $D_{\mathcal{S}}$ measures divergence from the baseline distribution to the masked distribution. We add 10^{-12} inside the baseline-probability logarithm for numerical stability.

We also record the answer-flip indicator,

$$f_{\mathcal{S}} = \mathbf{1}[\hat{c}_{\text{base}} \neq \hat{c}_{\mathcal{S}}], \quad (9)$$

where $\hat{c}_{\text{base}}$ and $\hat{c}_{\mathcal{S}}$ are the baseline and masked option predictions under the boundary experiment’s scoring rule in Sec. 4. The indicator $\mathbf{1}[\cdot]$ equals one when its condition holds and zero otherwise. We average divergence and answer flips across examples. Answer flips measure changes in the prediction, not changes in correctness against the ground-truth answer.

The single-layer profile shows where masking has the largest effect. Prefix masks test the accumulated effect of blocking early layers, while suffix masks test the need for direct visual access from a candidate layer onward. We inspect the suffix divergence and answer-flip curves to identify where this dependence becomes small. The experiment does not apply an automatic numerical threshold to select a boundary. This analysis guides the AVR schedule in Sec. 3.5. A small suffix effect concerns continued direct access to image tokens; it does not imply that later layers ignore visual information already carried by text-token representations. The measured profiles and the boundary appear in Sec. 5.1; the full tables, including the 36-layer checkpoint, are in Appendix E.

3.4 DWA : DEPTH WEIGHTED ATTENTION

DWA selects a region using information from the full model layer depth (Fig. 1). We first encode the full image toward 300 visual tokens and run the model with the complete question prompt. At each

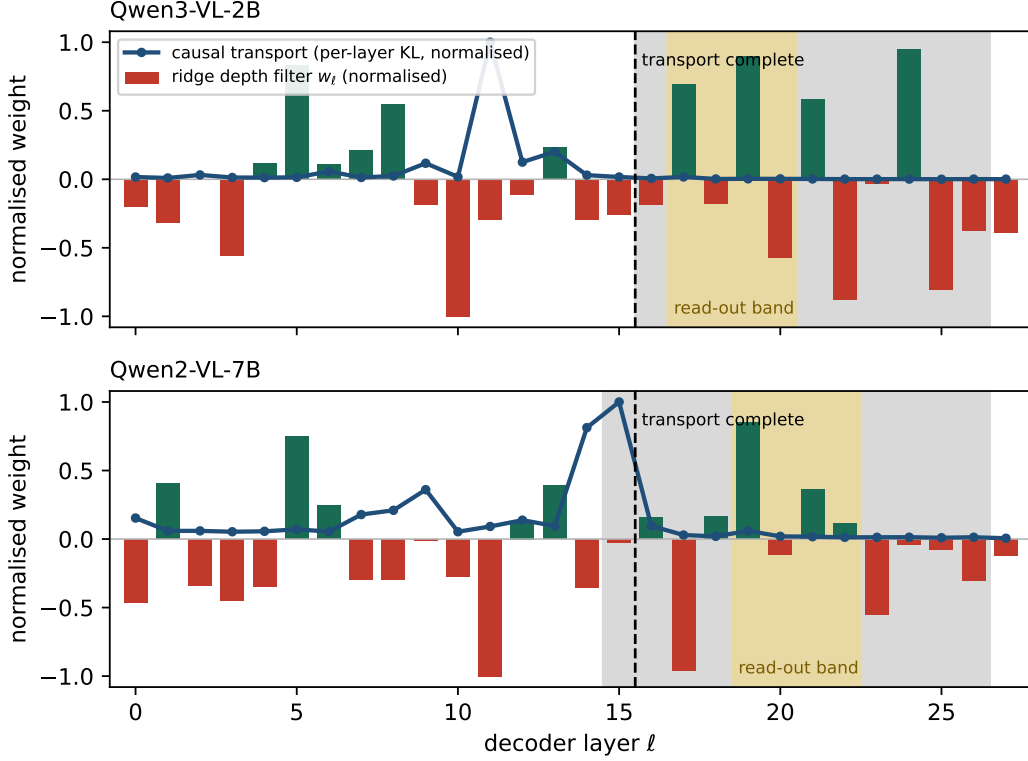


Figure 3: **Fitted depth weights and output sensitivity.** Normalised DWA log-attention coefficients and normalised single-layer masking KL for the two core checkpoints. The marked boundary indicates an approximate transition in output sensitivity. The bars show fitted log-attention coefficients; rank and spatial coefficients are omitted.

layer, we extract the final prompt position’s attention to image tokens and average it across attention heads.

Fig. 3 shows the fitted log-attention weights alongside the single-layer masking response.

Representing each image cell. Let $a_i^{(\ell)}$ denote the head-mean attention at image cell i and layer ℓ , where $i = 1, \dots, N$ and $\ell = 0, \dots, L - 1$, read from the final prompt token over the image cells. We normalise each map over the image cells:

$$A_i^{(\ell)} = \frac{a_i^{(\ell)}}{\max\left(\sum_{j=1}^N a_j^{(\ell)}, \epsilon\right)}, \quad \epsilon = 10^{-12}, \quad (10)$$

where j indexes image cells in the sum and ϵ prevents division by zero. Each cell is represented by its normalised depth profile,

$$\mathbf{x}_i = (\log(A_i^{(\ell)} + \epsilon))_{\ell=0}^{L-1} \in \mathbb{R}^L, \quad (11)$$

so with $L = 28$ each cell has 28 features. The read-out is a single closed-form ridge regression on these features,

$$w = (X^\top X + \alpha I)^{-1} X^\top y, \quad \alpha = 1, \quad (12)$$

with an unpenalised intercept, to the coverage label y_i , the fraction of the ground-truth box inside the $W=0.25$ window centred at cell i . Placement is the ring-masked arg-max of the score $s(c) = w_0 + \sum_{\ell=0}^{L-1} w_\ell \log A_\ell(c)$.

The feature engineered read-out additionally carries rank, neighbourhood and geometry features; they are specified in Ablations Section B, and a leave-one-group-in ablation shows the log-attention block alone carries the advantage.

Learning crop scores. The supervision target measures how much of the annotated evidence a crop would cover:

$$r_i = \frac{\text{area}(C_i \cap B)}{\max(\text{area}(B), \epsilon)}. \quad (13)$$

Here, C_i is the candidate crop centred at cell i , after shifting it inside the image boundaries. The box B is the smallest axis-aligned rectangle enclosing all annotated boxes for that example. The operator area measures area in normalised image coordinates, and r_i is the fraction of B covered by the crop. This is box coverage, rather than intersection over union.

We standardise each feature using the training-fold mean and standard deviation, adding 10^{-9} to the standard deviation. Ridge regression (Hoerl & Kennard, 1970) then fits the crop-coverage target:

$$(\mathbf{w}^*, \beta^*) = \arg \min_{\mathbf{w}, \beta} \{ \|\mathbf{X}\mathbf{w} + \beta\mathbf{1} - \mathbf{r}\|_2^2 + \lambda \|\mathbf{w}\|_2^2 \}, \quad \lambda = 1. \quad (14)$$

Here, $\mathbf{X} \in \mathbb{R}^{M \times (2L+7)}$ contains the standardised features of the M training cells, and $\mathbf{r} \in \mathbb{R}^M$ contains their coverage targets. The vector $\mathbf{w}$ contains the regression coefficients, β is an unpenalised intercept, and $\mathbf{1}$ is a length- M vector of ones. The notation $\|\cdot\|_2^2$ denotes the sum of squared entries, and $\arg \min$ returns the fitted parameters, marked by a star. The coefficient penalty has fixed strength $\lambda = 1$. We fit the regression coefficients and intercept by solving the corresponding linear system directly.

The evaluation uses five-fold splits grouped by example, so all cells from one example stay in the same fold. The procedure is repeated with seeds 700, 701, and 702, and the three held-out scores are averaged for each cell. Standardisation and coefficient fitting use only the training portion of each split.

Selecting and re-encoding the crop. We select the eligible cell with the highest averaged score. The outer grid ring is excluded when both grid dimensions exceed two cells; otherwise, every cell is eligible. The crop spans 25% of the original image width and 25% of its height. It is shifted inward when necessary to remain within the image, and its pixel coordinates are converted to integers. The crop is then re-encoded toward 300 visual tokens and used with the original question prompt for answering.

3.5 ALLOCATING VISUAL COMPUTATION WITH AVR

AVR examines whether later visual-token processing can be reduced while retaining higher-resolution input in the earlier layers. The main configuration encodes the full image toward 900 visual tokens, retains unrestricted processing through layer 16, and restricts visual access from layer 17 onward. No crop is applied.

Selecting tokens for continued access. For a final unrestricted layer b , visual tokens are ranked using

$$s_i^{\text{AVR}} = \frac{1}{b - \max(0, b - 4) + 1} \sum_{\ell=\max(0, b-4)}^b A_i^{(\ell)}. \quad (15)$$

Here, s_i^{AVR} is the selection score for image token i , and $A_i^{(\ell)}$ is the normalised, head-averaged attention from the final prompt position, computed on the higher-resolution input. With $b = 16$, the score averages layers 12–16. We retain the highest-scoring tokens, with the count given by

$$N_{\text{keep}} = \max(1, \text{round}(\rho N)), \quad \rho = 0.10. \quad (16)$$

Here, N is the realised input visual-token count, N_{keep} is the retained count, ρ is the retained fraction, and round denotes rounding to the nearest integer, with ties rounded to the nearest even integer. Additional configurations use layer-16-only attention or random selection at the same retained fraction.

Evaluating the restriction. The evaluation first collects attention in an unmasked forward pass. It then runs the model again with an additive bias of -10^4 on discarded visual-key columns in layers $b + 1, \dots, L - 1$. The bias applies to all query positions. It suppresses access to those visual keys while leaving the token tensors at their original lengths.

Table 1: Evaluation sets. Counts refer to the full sets; the core Qwen3-VL-2B placement comparison uses the 190 items available for every baseline. The layer sweep and architecture comparison use all 191 V*Bench items.

BENCHMARK	ITEMS	REPORTED PARTITION OR USE
V*Bench (Wu & Xie, 2024)	191	115 single-instance; 76 cross-instance
HR-Bench 4K (Wang et al., 2024b)	800	400 single-instance; 400 cross-instance
HR-Bench 8K (Wang et al., 2024b)	800	Resolution analysis; 400 per stratum
CV-Bench (Tong et al., 2024)	2,638	Count, relation, distance, and depth tasks
RealworldQA (xAI, 2024)	765	Multiple-choice and open-answer questions

The input-token target, retained fraction, and cutoff are fixed for each evaluated configuration; the main experiment does not use a per-example rule to choose the input resolution.

4 EXPERIMENTAL SETUP

Models and benchmarks. The core comparison uses Qwen3-VL-2B-Instruct (Qwen Team, 2025b) and Qwen2-VL-7B-Instruct (Wang et al., 2024a). Additional V*Bench comparisons use Gemma-3-4B (Gemma Team, 2025), InternVL3-8B (Zhu et al., 2025), and LLaVA-OneVision-7B (Li et al., 2024). The resolution analysis also includes Qwen2.5-VL-7B (Qwen Team, 2025a) and Qwen3-VL-8B. Table 1 summarises the evaluation sets. We retain the supplied question categories; RealworldQA’s answer-format split is not treated as an annotation of whether a question requires local or scene-level evidence.

Comparisons and transfer. We compare DWA with an uncropped image at the reference budget, a fixed-layer crop at layer 14, and a block-mean attention crop. The latter two are placement controls within the same crop-only pipeline, rather than reproductions of the complete ViCrop method (Zhang et al., 2023). The LASER-based control (Zhu et al., 2026) uses contrastive attention and per-example layer selection, but retains our quarter-width crop and omits LASER’s contrastive decoding. It selects the layer with the largest Euclidean norm of the positive difference between normalised head-mean query and no-query maps. The no-query map requires an additional 300-token pass, which is excluded from the nominal 600-token placement accounting. Ground-truth-guided crops serve as an oracle-placement reference, not a deployable method.

For cross-benchmark evaluations, ridge coefficients are fitted on V*Bench and transferred within the same checkpoint without target bounding-box supervision. The dedicated HR-Bench transfer protocol standardises features over the unlabelled target set; the extended benchmark protocol standardises them within each image. These transfer runs are reported separately from the grouped, out-of-fold V*Bench comparison. The extended AVR runs use random retention of 10% of visual tokens, with the first restricted layer at round(0.57L); this differs from the attention-ranked core schedule in Sec. 3.5.

Statistical comparisons. We use paired item bootstrap intervals (Efron, 1979) with 8,000 resamples and report 95% confidence intervals for accuracy differences. All differences are in percentage points. We identify a difference as statistically significant when its interval excludes zero. The repeated grouped splits for DWA are described in Sec. 3.4.

Answer scoring. We evaluate DWA and AVR using next-token logits at the final prompt position. For each answer option, we combine the scores of the token IDs obtained from its plain and space-prefixed letter:

$$g_c = \log \sum_{k \in \mathcal{J}_c} \exp(z_k), \quad (17)$$

$$\pi_c = \frac{\exp(g_c)}{\sum_{c' \in \mathcal{C}} \exp(g_{c'})}, \quad \hat{c} = \arg \max_{c \in \mathcal{C}} \pi_c. \quad (18)$$

Here, $\mathcal{C} = \{A, B, C, D\}$ is the set of answer options, indexed by c and c' . The set $\mathcal{J}_c$ contains the distinct final token IDs returned when tokenising the plain and space-prefixed forms of letter c without special tokens. The quantity z_k is the logit for vocabulary token k , g_c is the combined option score, π_c is its scalar probability after normalisation over the four options, and $\hat{c}$ is the predicted option.

The depth-masking experiment instead uses one token ID per option: the first ID returned when the plain letter is encoded without special tokens. Its answer prediction is the option with the largest logit among those four IDs. Its primary divergence measure covers the full vocabulary; it also records four-option divergence for the all-layer and single-layer interventions.

For CV-Bench and the multiple-choice portion of RealworldQA, the option-letter score is evaluated over the supplied choices and also includes the final token ID of the opening-parenthesis form, such as (A. Open-answer RealworldQA items use greedy generation of at most eight tokens and exact matching after lowercasing and removing punctuation, articles, and extra whitespace.

Performance and data use. Answer accuracy measures agreement between the predicted option and the ground-truth label. Crop localisation is evaluated using the box coverage defined in Sec. 3.4. Where a binary coverage hit is reported, a crop must cover at least half of the enclosing box. We draw the boundary-analysis examples from V*Bench without a separate held-out split for boundary selection.

Resolution targets and computation accounting. We resize images with bicubic interpolation toward a target visual-token count and check the count using the model’s image processor. We allow up to six resizing attempts and keep the closest candidate. The relative stopping tolerance is 8% for the depth analysis and DWA, and 6% for AVR. Target counts therefore need not equal realised counts.

The reference full-image input targets 600 visual tokens. DWA’s budget includes both passes described in Sec. 3.4. For an AVR token-removal schedule, the visual token-layer proxy is

$$C_{\text{TL}} = N(b + 1) + N_{\text{keep}}(L - b - 1). \quad (19)$$

Here, C_{TL} counts visual tokens multiplied by the number of layers through which they would be processed. The first term represents full-width processing through layer b , and the second represents processing of retained tokens afterward. For exactly 900 input tokens, $b = 16$, $L = 28$, and $N_{\text{keep}} = 90$, this gives 16,290 visual token-layers, compared with 16,800 for exactly 600 tokens across all 28 layers. We compute this proxy using the realised token counts.

This proxy quantifies the visual-token processing of the specified token-removal schedule. Vision-encoder and text-token processing lie outside its scope. The masking evaluation in Sec. 3.5 measures the schedule’s effect on accuracy while retaining full-length token tensors; C_{TL} is the cost assigned to physical token removal.

5 RESULTS

5.1 VISUAL ACCESS BECOMES LESS INFLUENTIAL AT LATER DEPTHS

The interventions in Fig. 2 show that continued direct image access has a smaller effect near the end of the decoder. Single-layer masking produces its largest output change at layer 11 in Qwen3-VL-2B and layers 14–15 in Qwen2-VL-7B. Masking the suffix beginning at layer 16 gives mean KL divergences of 0.014 and 0.029, compared with 1.266 and 0.532 when all layers are masked. Answer changes decrease to approximately 12% and 5% at that cutoff and to zero from a layer-24 suffix on both checkpoints. These profiles locate the transition from high to low sensitivity to continued direct visual access. Fig. 4 shows the same structure on real items: the attention map changes character at the boundary, and masking one layer at a time moves the output only before it.

Table 2 gives the full prefix/suffix grid, including the 36-layer checkpoint.

The scope of masking also matters. Restricting only the final prompt position across all layers gives KL divergences of 0.025 and 0.034, with 2.5% and 0% answer changes. Restricting all post-image text positions instead changes approximately 42% and 45% of answers. This contrast supports

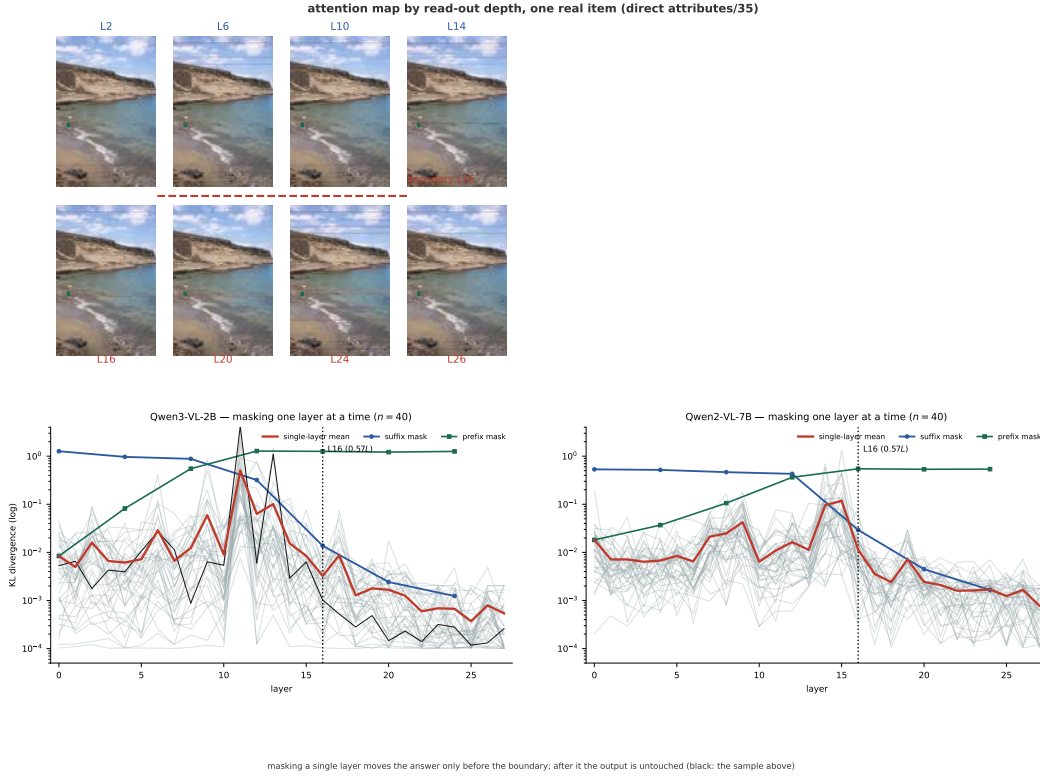


Figure 4: **The boundary on real items.** Top: one V*Bench item’s attention map at selected read-out depths, pre-boundary above and post-boundary below the dashed line; the late maps collapse onto a row-boundary band. Bottom: single-layer masking KL for all 40 calibrated items (thin), the mean (bold), and the prefix/suffix schedules, both models, with the boundary at L16 (0.57L). Black: the sample shown above.

Table 2: Prefix and suffix masks, all text rows, $n = 40$ per model, V*Bench at 300 tokens. The prefix mask $L_{0..\ell}$ saturates at the full-mask value once ℓ passes the window; the suffix mask $L_{\ell..end}$ falls to zero once ℓ is past it. Both read the boundary at the same place, and the 36-layer checkpoint puts it at the same fraction of depth ($\ell=20$ of 36).

	$\ell=4$	$\ell=8$	$\ell=12$	$\ell=16$	$\ell=20$	$\ell=24$
Qwen3 prefix $L_{0..\ell}$ (KL)	0.082	0.549	1.278	1.259	1.215	1.257
Qwen3 suffix $L_{\ell..}$ (KL)	0.967	0.880	0.320	0.014	0.002	0.001
Qwen3 suffix (answers flipped)	53%	50%	20%	12%	0%	0%
Qwen2 prefix $L_{0..\ell}$ (KL)	0.037	0.105	0.364	0.547	0.534	0.538
Qwen2 suffix $L_{\ell..}$ (KL)	0.518	0.465	0.429	0.029	0.004	0.002
Qwen2 suffix (answers flipped)	53%	47%	40%	5%	3%	0%
Qwen3-8B (36L) prefix $L_{0..\ell}$ (KL)	0.100	0.309	1.190	1.740	1.559	1.565
Qwen3-8B (36L) suffix $L_{\ell..}$ (KL)	1.671	1.538	1.278	0.413	0.052	0.009
Qwen3-8B (36L) suffix (answers flipped)	35%	45%	40%	15%	5%	0%

an indirect route through preceding text representations and explains why attention at the answer position alone need not identify where visual information influenced the prediction.

The boundary is a fraction of depth rather than a layer index: on a 36-layer checkpoint (Qwen3-VL-8B) under the same mask, the suffix is already nearly harmless at L20 (KL 0.052, 5% flips) and fully harmless from L24, and pruning at L21 — the same 0.57 fraction — is free, while a cut at $0.14 \times$ depth costs 7.9 points.

Table 3: Mask scope at all layers. KL divergence and answer flips, both core checkpoints ($n = 40$ each).

mask scope (all layers)	Qwen3-VL-2B	Qwen2-VL-7B
answer-emission token’s row only	0.025 / 2.5%	0.034 / 0%
all text rows after the image	1.266 / 42%	0.532 / 45%

Table 4: Late-depth agnosticism. KL divergence and answer flips for three probes, both core checkpoints. The all-post-image row is the full-text scope for reference.

probe	Qwen3-VL-2B	Qwen2-VL-7B
suffix mask @L16 (KL / flips)	0.014 / 12%	0.029 / 5%
suffix mask @L20 (KL / flips)	0.002 / 0%	0.004 / 2%
suffix mask @L24 (KL / flips)	0.001 / 0%	0.002 / 0%
image-value patch @L16 (KL / flips)	0.009 / 3.3%	0.005 / 5.0%
image-value patch @L20 (KL / flips)	0.0005 / 0%	0.0008 / 1.7%
answer-row mask, all layers (KL / flips)	0.025 / 2.5%	0.034 / 0%
all-post-image mask, all layers (KL / flips)	1.266 / 42%	0.532 / 45%

Table 3 compares the two mask scopes directly.

Three further properties of the late stack follow from these measurements and are detailed in Appendix G. First, the late layers are *agnostic to the image*: a suffix mask at layer 16 costs 0.014/0.029 KL with 12%/5% answer changes and nothing at all from layer 24, replacing the image tokens’ values changes the output no more than irrelevant early text (0.009/0.005 at L16, 0.0005/0.0008 at L20), and masking only the answer-emission token’s image access changes 2.5%/0% of answers. Second, *image and text tokens merge* in the late stack: the patching controls invert across depth (at L4 the image matters $151 \times / 131 \times$ more than text; at L20 the text matters $2492 \times / 965 \times$ more than the image), and the answer position reads the image only through the text representations that consolidated it. Third, the *layers disagree* about where to look: mean arg-max agreement across layer pairs is 13.8%/38.1%, the agreement matrix splits into two clusters at the boundary, and only the layers after it carry a usable placement signal (coverage of the evidence box is at most 20%/5% before the boundary, 41–56%/33–53% just after it, decaying toward the output). Reading attention at a depth that carries the image is therefore not the same as reading it where placement is decodable, and the two occupy disjoint ranges. Tables 4 and 5 quantify the first and third properties.

5.2 DWA IMPROVES CROP SELECTION

Table 7 compares placement rules with the image budget and crop size held fixed. DWA reaches 72.6% and 70.2% accuracy, compared with 64.7% and 61.3% for the strongest evaluated placement baseline, the LASER-based control (Zhu et al., 2026). The paired gains are +7.9 [+1.1, +14.7] and +8.9 [+3.1, +14.7] points. On single-instance questions, DWA reaches 77.2% and 71.3%, while its cross-instance accuracies are 65.8% and 68.4%. DWA therefore improves crop selection over both fixed and adaptive single-layer read-outs within the common placement protocol.

The localisation measurements show a corresponding improvement. With the outer grid ring excluded for both methods, DWA covers at least half of the annotated evidence box on 63.9% and 55.0% of examples, compared with 46.6% and 39.3% for the block mean. Fig. 5 illustrates cases where this change in placement also corrects the answer.

Comparison with native resolution. Table 8 places these gains against the models’ native dynamic-resolution inputs. DWA narrows the accuracy gap to 3.7 and 1.0 points using approximately 18% and 14% of the native median visual-token counts. Both paired intervals include zero. AVR reaches 70.0% and 61.8%, compared with the native reference of 76.3% and 71.2%; the Qwen2-VL-7B difference is significant. Oracle crops exceed native accuracy by 11.6 and 16.2 points, confirming the value of precise placement even against higher-resolution full-image input.

Table 5: Layer disagreement and the read-out band. Agreement over layer pairs; coverage of the evidence box by a single-layer crop ($\geq 50\%$ of the box), by read-out depth.

	quantity	Qwen3-VL-2B	Qwen2-VL-7B
	mean arg-max agreement (28×28 pairs)	13.8%	38.1%
	coverage before the boundary (L0–L15)	$\leq 20\%$	$\leq 5\%$
	coverage just after (L17–L22)	41–56%	33–53%
	coverage near the output (L25–L27)	3–28%	5–9%

Figure 5: **Crop placement and answer changes on Qwen3-VL-2B.** The panels show the full image, block-mean attention, held-out DWA scores, and the two resulting crops. Both rules exclude the outer grid ring. These are the first three question IDs among nine qualifying single-instance examples where DWA covers at least half of the evidence box and answers correctly, the block mean does neither, and the recomputed DWA location matches the recorded run. They illustrate successful corrections under this selection criterion.

5.3 AVR CONVERTS THE BOUNDARY INTO RESOLUTION

AVR removes visual tokens after the boundary and spends the saving on higher-resolution encoding. Two controls separate the two components. Against the unmodified model at the same 900-token input (`uniform@900`), the later restriction is within 0.5 points in seven of the ten benchmark-checkpoint-stratum cells, two cells are positive by 1.3 and 2.6 points, and one (HR-Bench 8K cross-instance) shows a significant loss of 1.8 points; at the boundary the choice of retained tokens is irrelevant, so random keep matches attention keep at every keep ratio and the pruning itself is free. What remains is the resolution: AVR’s gain over the equal-compute bar tracks the accuracy the checkpoint already gains from 900 over 600 tokens, at slope 1.01 with $r = 0.966$ across ten cells and four checkpoints (Table 9). The identity predicts where AVR fails: the one cell with a negative point estimate, HR-4K cross-instance, is the cell whose baseline is itself worse at 900 tokens than at 600.

The risk profile follows from the identity. AVR is positive on both strata in 6 of 7 cells and never significantly worse; the gains concentrate where headroom exists. On HR-4K and HR-8K it improves single-instance accuracy by +4.0/+2.8 and +6.5 points, and on V*Bench it is worth +10.5 on cross-instance for Qwen3-VL-2B — the stratum where cropping loses. Because it never discards coverage, AVR is the policy of choice where the evidence is relational or scene-level; because

Table 6: DWA accuracy across all four benchmarks and all four models. *Reference* is the uncropped image at the equal-compute budget (600 visual tokens; the model’s native encoding where it has no resolution ladder). Differences are paired and in percentage points; ✓ = interval excludes zero.

Model	Benchmark	n	Reference	DWA	DWA – reference
Qwen3-VL-2B	V*	191	62.3	72.8	+10.5 ✓
Qwen3-VL-2B	HR-4K	800	59.0	62.5	+3.5
Qwen3-VL-2B	CV-Bench	2,638	79.9	65.8	−14.1 ×
Qwen3-VL-2B	RealworldQA	765	62.6	56.9	−5.8 ×
Qwen2-VL-7B	V*	191	59.2	68.1	+8.9 ✓
Qwen2-VL-7B	HR-4K	800	57.0	59.1	+2.1
Qwen2-VL-7B	CV-Bench	2,638	75.0	66.0	−9.0 ×
Qwen2-VL-7B	RealworldQA	765	64.7	57.5	−7.2 ×
LLaVA-OV-7B	V*	191	51.8	65.4	+13.6 ✓
LLaVA-OV-7B	HR-4K	800	48.2	56.5	+8.2 ✓
LLaVA-OV-7B	CV-Bench	2,638	76.6	65.9	−10.7 ×
LLaVA-OV-7B	RealworldQA	765	61.6	59.2	−2.4
InternVL3-8B	V*	191	74.9	75.9	+1.0
InternVL3-8B	HR-4K	800	70.5	65.6	−4.9 ×
InternVL3-8B	CV-Bench	2,638	81.7	68.6	−13.1 ×
InternVL3-8B	RealworldQA	765	67.6	59.9	−7.7 ×

Table 7: V*Bench accuracy (%) under the common placement protocol. Qwen3-VL-2B uses 190 matched items and Qwen2-VL-7B uses 191. Single and cross denote single-instance and cross-instance questions. The full-image reference uses 600 tokens; DWA uses 300 for localisation and 300 for answering. All crop controls answer at 300 tokens. The fixed-layer, block-mean, and LASER-based rows are placement controls as specified in Sec. 4; the LASER-based control also requires a no-query pass. The oracle row uses ground-truth localisation.

Method	Qwen3-VL-2B			Qwen2-VL-7B		
	Single	Cross	All	Single	Cross	All
Full image, 600 tokens	62.3	65.8	63.7	57.4	59.2	58.1
Fixed layer 14	24.6	57.9	37.9	33.9	44.7	38.2
Block-mean crop	64.0	59.2	62.1	67.0	47.4	59.2
LASER-based placement	64.0	65.8	64.7	65.2	55.3	61.3
DWA	77.2	65.8	72.6	71.3	68.4	70.2
Oracle placement	96.5	75.0	87.9	93.9	77.6	87.4

it requires a resolution ladder, it is undefined for checkpoints that quantise resolution (InternVL3, Gemma-3), which we report rather than fake. Table 10 gives AVR’s accuracy across every benchmark and model, and Table 11 the full risk profile.

5.4 THE BENEFIT OF CROPPING DEPENDS ON THE REQUIRED EVIDENCE

The transferred evaluations in Table 12 separate the effect of cropping on local and scene-level evidence. On HR-Bench 4K, DWA improves single-instance accuracy by 14.5 and 11.0 points, but reduces cross-instance accuracy by 7.5 and 6.8 points. The pooled gains are 3.5 and 2.1 points. On CV-Bench, the losses are larger: 14.1 and 9.0 points. The pooled RealworldQA differences are also negative, at −5.8 and −7.2 points.

This pattern supports using concentrated resolution for local evidence and preserving scene coverage for counting or comparing separated regions.

AVR outperforms DWA on both cross-instance and CV-Bench comparisons for each checkpoint, with all four paired intervals excluding zero (Table 12). AVR’s full-image allocation therefore better preserves performance on tasks requiring evidence from multiple regions. DWA concentrates resolution spatially, while AVR allocates visual access across depth.

Table 8: V*Bench accuracy (%) against native dynamic resolution on the same matched items as Table 7. Native median visual-token counts are 3,290 and 4,320, respectively. DWA targets 600 tokens over two passes, approximately 18% and 14% of those medians. AVR targets 900 input tokens with the later restriction in Sec. 3.5; its nominal token-layer proxy is 16,290. Token accounting follows Sec. 4.

Method	Qwen3-VL-2B	Qwen2-VL-7B
Native dynamic resolution	76.3	71.2
Full image, 600 tokens	63.7	58.1
Block-mean crop	62.1	59.2
LASER-based placement	64.7	61.3
AVR	70.0	61.8
DWA	72.6	70.2
Oracle placement	87.9	87.4

Table 9: AVR’s gain against the headroom $\text{acc}(900) - \text{acc}(600)$, for the ten stratum-cells that carry the `uniform@900` diagnostic. Pearson $r = 0.966$, OLS slope 1.01, mean $|\text{gain} - \text{headroom}| = 0.7$ points. Headroom is measurable from two baseline runs with no method involved.

cell	stratum	AVR – reference	headroom
V*Bench Qwen2.5-VL-7B	single	+12.2 ✓	+12.2
	cross	+7.9	+5.2
V*Bench Qwen3-VL-8B	single	+7.0	+7.0
	cross	+7.9 ✓	+6.6
HR-4K Qwen3-VL-2B	single	+4.0 ✓	+4.0
	cross	−2.8	−3.0
HR-4K Qwen2-VL-7B	single	+2.8	+2.8
	cross	+0.8	+0.2
HR-8K Qwen3-VL-2B	single	+6.5 ✓	+6.8
	cross	+1.8	+3.5

Read-out improvements across architectures. Table 13 separates two comparisons: replacing the block-mean read-out at the same cost, and replacing a full-image pass given the crop pipeline’s total token budget. DWA exceeds the block mean on all five checkpoints, with paired intervals excluding zero on four. Its advantage over the budget-matched full-image reference is significant on the two Qwen checkpoints and Gemma-3. InternVL3 and LLaVA-OneVision show positive point estimates of 5.8 and 1.0 points, with intervals including zero. These model-specific fits show that the improvement over a block-mean read-out extends across architectures.

6 DOES THE DEPTH BOUNDARY GENERALISE BEYOND VISION?

The boundary in Section 5.1 is a claim about where a language model stops reading a non-text modality, not about images in particular. We therefore applied the same masking protocol, unchanged, to an audio–language model and a video–language model, and then asked whether the two policies follow. Each modality is represented by a single checkpoint, so this section reports transfer of a measurement, not a multi-model claim; the results in Section 5 remain the basis for the paper’s conclusions.

6.1 THE BOUNDARY IS MODALITY-GENERAL; ITS DEPTH IS NOT

Table 14 reports the protocol applied to three modalities. In each case masking every text row’s attention to the modality columns from layer ℓ onward leaves the output unchanged once ℓ passes a fixed depth, and the all-layer control moves the output substantially.

Two observations follow. First, transport completes at a well-defined depth in every modality tested: the phenomenon is a property of how these backbones consume non-text tokens, not of images. Second, the completion depth is not constant. Audio completes at 0.75 rather than 0.57, which falsified the depth we had registered in advance for that modality. The depth of peak single-layer

Table 10: AVR accuracy across all four benchmarks and all four models. *Reference* is the uncropped image at the equal-compute budget (600 visual tokens). Differences are paired and in percentage points; ✓ = interval excludes zero. AVR is infeasible for InternVL3-8B, which has no resolution ladder.

Model	Benchmark	n	Reference	AVR	AVR – reference
Qwen3-VL-2B	V*	191	62.3	67.5	+5.2 ✓
Qwen3-VL-2B	HR-4K	800	59.0	59.9	+0.9
Qwen3-VL-2B	CV-Bench	2,638	79.9	80.4	+0.5
Qwen3-VL-2B	RealworldQA	765	62.6	63.3	+0.7
Qwen2-VL-7B	V*	191	59.2	64.9	+5.8 ✓
Qwen2-VL-7B	HR-4K	800	57.0	57.1	+0.1
Qwen2-VL-7B	CV-Bench	2,638	75.0	71.5	-3.5 ✗
Qwen2-VL-7B	RealworldQA	765	64.7	62.6	-2.1
LLaVA-OV-7B	V*	191	51.8	58.6	+6.8 ✓
LLaVA-OV-7B	HR-4K	800	48.2	46.2	-2.0
LLaVA-OV-7B	CV-Bench	2,638	76.6	76.0	-0.6
LLaVA-OV-7B	RealworldQA	765	61.6	60.0	-1.6
InternVL3-8B	(all)	—	—	—	—

Table 11: Risk profiles at the equal-compute reference: AVR and the crop, per stratum, seven cells. ✓ = paired interval excludes zero.

cell	AVR single	AVR cross	crop single	crop cross
V* Qwen3-VL-2B	+3.5	+10.5 ✓	+14.8 ✓	+0.0
V* Qwen2-VL-7B	+3.5	+1.3	+13.9 ✓	+9.2
V* Qwen2.5-VL-7B	+12.2 ✓	+7.9	+12.2 ✓	-2.6
V* Qwen3-VL-8B	+7.0	+7.9 ✓	+6.1	-3.9
HR-4K Qwen3-VL-2B	+4.0 ✓	-2.8	+10.2 ✓	-8.8 ✗
HR-4K Qwen2-VL-7B	+2.8	+0.8	+9.2 ✓	-6.0 ✗
HR-8K Qwen3-VL-2B	+6.5 ✓	+1.8	+8.8 ✓	-8.8 ✗

sensitivity, by contrast, is nearly invariant at 0.38–0.43. The layer at which the model reads a modality most strongly is stable; the layer by which it has finished reading is not.

6.2 WHAT AVR REQUIRES: AN INFORMATION KNOB, NOT A TOKEN KNOB

Section 5.3 established that AVR’s gain equals the accuracy difference between its high- and low-budget endpoints. Read as a precondition rather than a diagnostic, this identity states when AVR can

Table 12: Accuracy differences from the uncropped reference in the extended evaluation (percentage points). DWA coefficients are transferred from V*Bench. AVR uses the extended random-retention schedule. Each HR-4K stratum has 400 items; CV-Bench uses all 2,638 items. Dashes indicate values not included in this comparison. The paired AVR–DWA differences on HR-4K cross-instance and CV-Bench exclude zero for both checkpoints.

Model	Evaluation set	DWA – reference	AVR – reference
Qwen3-VL-2B	HR-4K single	+14.5	—
	HR-4K cross	-7.5	-1.5
	CV-Bench, all	-14.1	+0.5
Qwen2-VL-7B	HR-4K single	+11.0	—
	HR-4K cross	-6.8	-1.2
	CV-Bench, all	-9.0	-3.5

Table 13: Architecture comparison on all 191 V*Bench items. Accuracy and differences are in percent and percentage points, respectively. The cost ratio is DWA’s total visual tokens divided by the original full-image reference’s tokens. The final column instead uses a full-image reference matched to DWA’s total budget. An asterisk marks a paired 95% interval excluding zero. Qwen3 values differ slightly from Table 7 because that table excludes one item.

Model	DWA	Cost ratio	DWA – block	DWA – matched
Qwen3-VL-2B	72.8	1.00×	+10.5*	+8.9*
Qwen2-VL-7B	70.2	1.00×	+11.0*	+12.0*
Gemma-3-4B	52.9	2.00×	+16.2*	+17.3*
InternVL3-8B	76.4	8.37×	+10.5*	+5.8
LLaVA-OneVision-7B	67.0	2.27×	+5.2	+1.0

Table 14: The masking protocol of Section 3.3 applied unchanged to three modalities. “Completion” is the shallowest suffix mask whose mean output KL falls below 0.03; “peak” is the depth of the largest single-layer KL. Depths are fractions of the backbone’s layer count. Audio and video use the modality-used subset, matching the convention used for vision. Each modality is one checkpoint.

Modality	Backbone	Layers	Completion	Peak
Image	Qwen3-VL-2B	28	0.57	0.39
Audio	Qwen2-Audio-7B	32	0.75	0.38
Video	Qwen3-VL-2B	28	0.57	0.43

pay at all: only when raising the budget restores information that the low budget destroyed. Table 15 applies that test.

On short audio clips the only available knob is time-stretching, which lengthens the token sequence without adding acoustic detail. Headroom is negative and AVR does not pay, exactly as the identity predicts. On long clips the encoder must compress, so encoding natively recovers information the compressed input lost, and headroom becomes positive. On video the frame count is a genuine resolution knob and headroom is again positive.

Positive headroom is necessary but not sufficient. AVR spends its saving on a larger budget, and the size of that saving is set by the boundary itself. With L layers, a cut at layer P , and a retained fraction k , the affordable multiplier over the reference budget is

$$\kappa = \frac{L}{P + k(L - P)}. \quad (20)$$

A deeper boundary leaves fewer layers over which to save. For video, $P/L = 0.57$ and $k = 0.1$ give $\kappa = 1.63$; for audio, $P/L = 0.75$ gives $\kappa = 1.29$. The video headroom of 4.9 points was measured across an eightfold span, and at the affordable $1.5\times$ the measured AVR gain is +3.4 points with an interval that includes zero — which matches the +3.2 points of headroom available at that same budget, with pruning itself costing +0.2 points. The policy collects what is available to it; the boundary determines how much that is. Equation 20 makes this quantitative rather than incidental: the same measurement that locates the read-out also bounds the budget a depth-aware policy can reinvest.

Table 15: Headroom, measured against the strongest equal-budget reference in each setting, and the AVR gain it predicts. An asterisk marks a 95% interval excluding zero. Intervals are clustered by source item. Video CIs are clustered by base video; audio uses the first 180 s of each clip.

Setting	Budget knob	Headroom	AVR – reference
Audio, 10 s clips	time-stretch	−3.8	−5.0
Audio, 180 s clips	native encoding	+2.3*	—
Video, TempCompass	frames 4 → 32	+4.9*	+3.4

Table 16: Placement ceilings. The ceiling compares a label-selected crop against a label-selected reference at the same budget, so it isolates placement from selection. “Incumbent” is the block-mean read-out against a random window.

Setting	Ceiling (bias-matched)	Incumbent – random
Audio, MMAU	−3.3	+5.0*
Video, TempCompass	+5.5*	−0.7

6.3 WHAT PLACEMENT REQUIRES: ROOM ABOVE THE REFERENCE

For DWA the corresponding precondition is that some crop is better than no crop. We test this with an oracle that selects among candidate windows using the label, which upper-bounds any label-free placer. The oracle must be compared against a reference that is selected the same way: choosing the best of K draws beats a single draw even when no draw is better on average. On video, this selection effect alone is worth 3.7 points of the 9.1 points an uncontrolled comparison reports.

The two modalities fail the precondition in opposite ways (Table 16). In audio the label-free statistic carries real signal, but the best window chosen with the label is still worse than not cropping: placement has nothing to reach, so no placer was fitted. In video there is genuine room, but the block-mean statistic is indistinguishable from choosing at random. Refitting the read-out of Section 3.4 on video, with folds held out by source video, reproduces the in-sample fit but does not generalise, and remains below a budget-matched reference. What fails in video is not the value of placement but the availability of a label-free signal for it.

6.4 WHERE THE POLICIES ARE NOT EXPRESSIBLE

Two architectural observations bound the scope further, and neither is an experimental outcome.

No widely used audio–language model presents a frequency axis to the language model. Whisper-derived encoders, which cover most current systems, fold the mel axis into the channel dimension, so the token sequence is indexed by time alone; query-based adapters pool to an unordered set. Frequency is to audio what colour is to images: present in the signal, absent from the token grid. Selection over a frequency axis is therefore not expressible, independently of whether it would help.

Vision–language–action models fix their input resolution. OpenVLA resizes every observation to 224×224 with patch-14 towers and emits exactly 256 visual tokens regardless of input, and SmolVLA pads to a fixed size. Without a resolution ladder there is no budget to reallocate, so AVR has no free parameter in these systems. The same property makes cropping automatically budget-matched, which removes the equal-compute confound that otherwise dominates such comparisons.

6.5 SUMMARY

The measurement transfers: a depth at which a modality stops being read exists in every modality we tested, and the depth of peak sensitivity is nearly constant across them. The policies transfer under stated conditions: AVR requires a knob that restores destroyed information, and the boundary depth bounds how much budget it can reinvest through Equation 20; placement requires both room above the reference and a label-free signal that finds it. These conditions were derived from the vision experiments and predicted each cross-modal outcome in advance, including the negative ones.

7 ABLATIONS

7.1 CHOICE OF READ-OUT DEPTH

We vary the attention layer while keeping the crop size, answering procedure, and visual-token budget fixed. Fig. 6 shows a 35.6-point accuracy range on Qwen3-VL-2B and a 26.7-point range on Qwen2-VL-7B. The best layers are 17 and 21, respectively. Every tested read-out before layer 16 performs below the uncropped reference on both checkpoints. This separates the depth at which

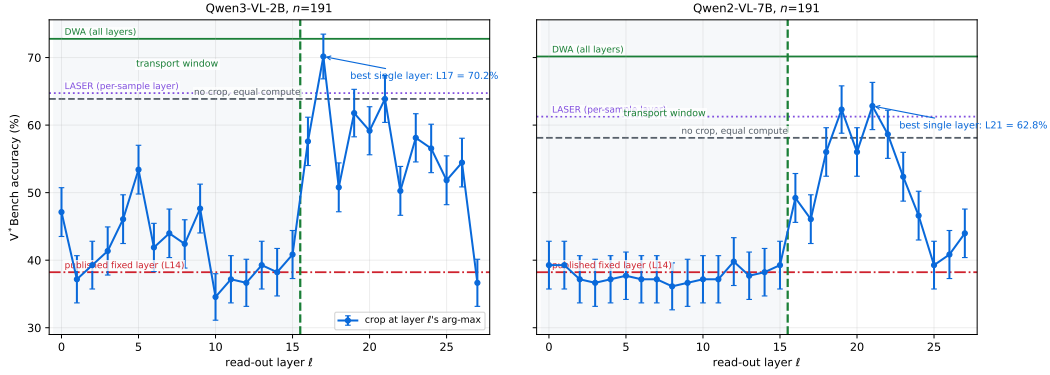


Figure 6: **Accuracy when only the crop read-out layer changes.** The layer sweep uses 191 V*Bench examples per checkpoint; error bars show one standard error. The fixed-layer and LASER (Zhu et al., 2026) lines denote the placement controls defined in Sec. 4. The dashed boundary is the approximate transition identified by masking.

image access affects the answer from the depth at which the final prompt position supplies a useful spatial map.

DWA exceeds the best single layer by 2.6 points on Qwen3-VL-2B and 7.3 points on Qwen2-VL-7B. Only the latter difference has a paired interval excluding zero. Since the best layer is selected using benchmark accuracy, this is an optimistic single-layer reference.

7.2 WHAT DOES THE FITTED READ-OUT CONTRIBUTE?

Weight signs. Constraining the 56 log-attention and rank coefficients to be non-negative produces no statistically significant end-task loss relative to DWA on either core checkpoint. The spatial coefficients and intercept remain unconstrained. The corresponding localisation fits set 44 and 43 of the 56 layer coefficients to zero, concentrating weight on a sparse subset of layers.

A training-free aggregation control. Taking the maximum over layers 17–20 on Qwen3-VL-2B and 19–22 on Qwen2-VL-7B gives 68.9% and 60.7% accuracy, compared with DWA’s 72.6% and 70.2% in the matched comparison. Fitting gives higher point accuracy on both checkpoints, with a significant advantage on Qwen2-VL-7B.

Fold assignment. Refitting with disjoint fold seeds changes pooled accuracy by at most one point on each checkpoint, with both paired intervals including zero. DWA is therefore stable across the tested grouped partitions.

7.3 AVR CUTOFF, RETAINED TOKENS, AND INPUT RESOLUTION

Cutoff and token ranking. Before the measured transition, attention-ranked retention exceeds random retention by 8.4 and 8.9 points in the reported comparison. At later cutoffs, the ranking advantage decreases. In the full Qwen3-VL-2B V*Bench run, attention and random retention give 70.2% and 68.6% accuracy and agree on 96.3% of answers. A schedule that begins reducing visual access within the sensitive depth range loses 1.0 and 5.8 points relative to the boundary-based schedule. These controls support retaining visual access through the sensitive range and applying the restriction after the transition.

Higher resolution versus restriction alone. We compare AVR with the unmodified model at the same 900-token input resolution. Across the ten evaluated benchmark-checkpoint-stratum combinations, seven accuracy differences are within 0.5 points. Two are positive by 1.3 and 2.6 points, while HR-Bench 8K cross-instance shows a significant loss of 1.8 points $[-3.2, -0.5]$. Later restriction thus keeps accuracy close to the unrestricted higher-resolution model across the evaluated settings.

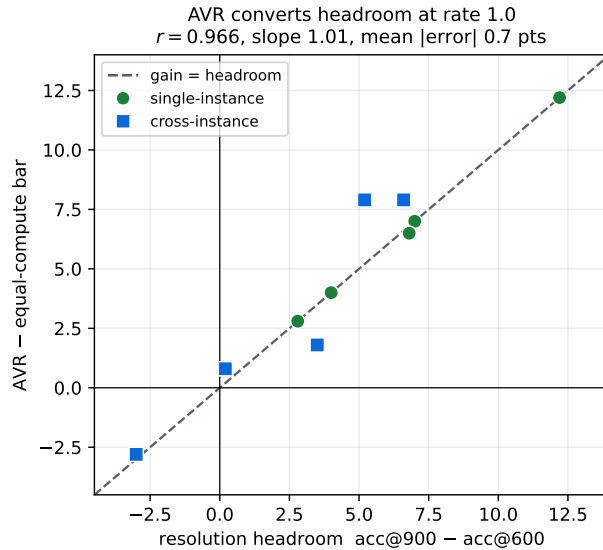


Figure 7: **AVR gain and the benefit of higher input resolution.** Each point is one of ten reported benchmark-checkpoint-stratum combinations with an unmodified 900-token reference. The horizontal axis measures its accuracy gain over 600 tokens; the vertical axis measures AVR’s gain over that same reference budget. The diagonal indicates equal gains. Both axes share the same 600-token reference.

Fig. 7 shows that AVR’s gain tracks the benefit of increasing input resolution, with a reported correlation of 0.966 and a mean absolute difference of 0.7 points. Together with the same-resolution control, this identifies additional input detail as the main source of AVR’s accuracy gain. The later restriction allocates the token-layer budget toward that detail.

8 CONCLUSION

Decoder depth determines both when direct image access influences prediction and where attention provides a useful spatial read-out. Our interventions and layer sweeps separate these roles. DWA uses fitted depth weights to improve crop placement, while AVR preserves full-image coverage and concentrates visual-token processing in the earlier decoder layers.

DWA improves local-detail answering, and its advantage over block-mean crop selection extends across the evaluated architectures. AVR better preserves performance on tasks requiring evidence from multiple regions. Together, these findings provide an empirical basis for choosing spatial concentration or full-image coverage according to the required evidence. Future work will optimize physical token removal and evaluate end-to-end latency and memory, alongside adaptive selection of the spatial allocation policy.

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AUTHOR CONTRIBUTIONS

If you'd like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

ACKNOWLEDGMENTS

Use unnumbered third level headings for the acknowledgments. All acknowledgments, including those to funding agencies, go at the end of the paper.

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A EXPERIMENTAL PROTOCOL

Models. Qwen3-VL-2B-Instruct and Qwen2-VL-7B-Instruct (core pair), Qwen2.5-VL-7B-Instruct, Qwen3-VL-8B-Instruct (36 layers), LLaVA-NeXT-7B and LLaVA-OneVision-7B (attention analyses only). `bfloat16`; eager attention wherever hooks or a prune bias are required, SDPA otherwise. We note a ~ 2 point kernel-dependent difference on the 7B between the two implementations and always compare arms computed under the same kernel.

Budgets. Images are resized so that the processor’s own `image_grid_thw` yields the target token count ($\pm 8\%$); realised counts are recorded per arm and a contrast is voided if budgets drift by more than 10%. Where pruning changes cost mid-network we report token-layers.

Read-out. The attention map is the final prompt token’s attention over image-token columns, averaged over heads *before* normalising (the order matters: normalising per head first up-weights sink-heavy heads and shifts the block-mean read-out by 11 points on the 7B). The prompt must end at the answer-emission point; a prompt ending elsewhere costs 36–42 points on the arg-max read-out.

Statistics. Paired item bootstrap with 8000 resamples. Learned components use GroupKFold(5) grouped by item $\times 3$ seeds; the headline DWA result is additionally reproduced under a disjoint fold-seed set (difference within ± 1 point). Run-to-run floor on coverage is 1–2.5 points.

Pipeline discipline. An exactly $+0.0$ contrast with a $[+0.0, +0.0]$ interval is treated as a pipeline fault, never a result; this rule caught a monkey-patch that silently failed to reach one model family. Every pruning run asserts at runtime that the loaded model’s attention module is the one that was patched.

B THE DWA READ-OUT: FEATURES, ESTIMATOR, AND THE FEATURE-BLOCK ABLATION

Let $a_i^{(\ell)}$ denote the head-mean attention at image cell i and layer ℓ , where $i = 1, \dots, N$ and $\ell = 0, \dots, L - 1$, read from the final prompt token over the image cells. We normalise each map over the image cells,

$$A_i^{(\ell)} = \frac{a_i^{(\ell)}}{\max\left(\sum_{j=1}^N a_j^{(\ell)}, \epsilon\right)}, \quad \epsilon = 10^{-12}, \quad (21)$$

where j indexes image cells in the sum. Each cell is represented by its normalised depth profile, $(\log(A_i^{(\ell)} + \epsilon))_{\ell=0}^{L-1}$, so with $L = 28$ each cell has 28 features. The read-out is a single closed-form ridge regression on these features,

$$w = (X^\top X + \alpha I)^{-1} X^\top y, \quad \alpha = 1, \quad (22)$$

with an unpenalised intercept, fitted out-of-fold (GroupKFold(5) $\times 3$ group permutations, grouped by item) to the coverage label y_i , the fraction of the ground-truth box inside the $W=0.25$ window centred at cell i . Placement is the ring-masked arg-max of the score $s(c) = w_0 + \sum_{\ell=0}^{L-1} w_\ell \log A_\ell(c)$.

The deployed vector. The deployed read-out carries two further blocks, $\mathbf{x}_i \in \mathbb{R}^{2L+7}$: (i) L per-layer rank features, $R_i^{(\ell)} = \text{rank}_\downarrow(a_i^{(\ell)}) / (N - 1)$, the cell’s descending rank in layer ℓ ’s map; and (ii) seven neighbourhood/geometry features — the $\log 3 \times 3$ neighbourhood mean of the read-out-band map (the band where attention first becomes question-conditioned, $L_{\text{lo}} = \text{round}(0.57L)$ through $L - 1$), the cell’s horizontal and vertical coordinates, its distance from the image centre, its distance to the nearest border, and two indicators for the last row and column. Ranks are per-layer and item-relative; the geometry block is item-independent.

Feature-block ablation. Refitting the same estimator on V*Bench (191 items) with feature blocks removed, out-of-fold coverage of the $W=0.25$ window at the ring-masked arg-max:

The log-attention block alone recovers 0.163/0.166 of the block-mean gap — 91%/108% of the full vector’s lift: the advantage is carried by the depth profile, and the rank/geometry blocks add $+0.017$ on Qwen3 and -0.013 on Qwen2. They are kept in the deployed vector because they are free at fit time, but they are not the mechanism; the main-text read-out quotes the log-attention score. Script: `phase238_feat_ablation.py`.

Table 17: **The log-attention block carries the advantage.** Out-of-fold coverage on V*Bench, by feature block.

model	block-mean read-out	log only (L)	log+rank ($2L$)	full ($2L+7$)
Qwen3-VL-2B	0.475	0.638	0.640	0.655
Qwen2-VL-7B	0.408	0.574	0.587	0.561

C INTERMEDIATE METRICS SYSTEMATICALLY OVER-REPORT

Five times in this work an intermediate metric improved while end-task accuracy did not: a read-out that closed the coverage gap but converted 3.1 points worse; a ridge variant that tied on coverage and differed on the chosen cell for two-thirds of items; an early-exit localiser free in coverage but -3.7 pooled at the end task; a higher-resolution localiser that gained $+4.2$ coverage and -3.7 accuracy; and an objectness probe that retains the evidence box far better than attention (87% vs. 72%) and converts to -1.6 . We therefore report coverage only as a diagnostic and gate every claim on end-task accuracy at matched compute.

D THE TRANSPORT MEASUREMENT IN FULL

A forward pre-hook on every decoder layer adds -10^4 to the attention logits of chosen query rows over the image-token columns, at chosen layers only. Masking *all* text rows at *all* layers destroys the answer (KL 1.266 / 0.532, 42% / 45% of answers flip, $n=40$); masking only the answer token’s row at all layers does not (KL 0.025, 3% flip). Table 2 localises the window.

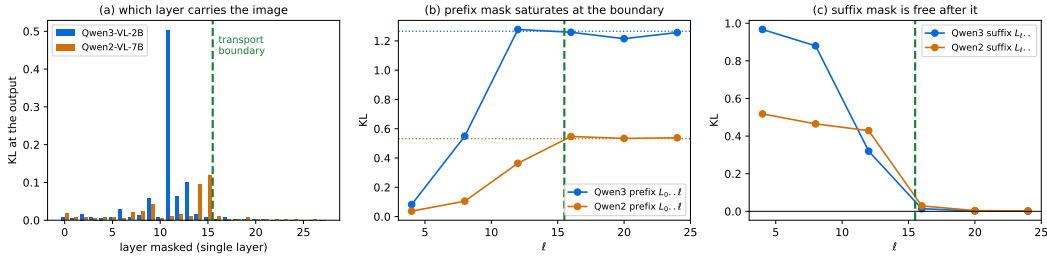


Figure 8: **The transport measurement.** (a) Masking one layer’s image attention at a time: the effect is concentrated well before the boundary and is at or below 0.0023 from L20 on both models. (b) Masking a prefix $L_{0..l}$ reaches the full-mask value (dotted) once l passes the window. (c) Masking a suffix $L_{l..end}$ costs nothing once l is past it. Three readings, one boundary.

Single-layer masks peak at L11 on Qwen3 (KL 0.504; next highest L13 at 0.101) and at L15/L14 on Qwen2 (0.118/0.096). From L20 onward every layer is at or below 0.0023 on both models. On Qwen3-VL-8B, which has 36 layers, the same behaviour appears at L21, i.e. the same 0.57 fraction of depth.

Three independent readings of the same boundary. The masks are one. Pruning is the second: discarding 90% of visual tokens at the boundary costs $+0.0$ to $+0.5$ points across ten stratum-cells, whereas moving the same cut to $l=4$ costs 5.2 more on one model (0.5 on the other), and a taper that removes half the tokens at $l=9-12$ loses 3.5 points against not pruning at matched tokens while buying nothing over a single cut (-1.0). Read-out depth is the third: restricting the localiser to layers $\leq L20$ costs -0.5 and $+1.6$ points of coverage, while restricting it to $\leq L16$ costs 9.4 and 22.0. Transport finishes at the boundary; the signal the ranking reads is produced after it.

Table 18: Where each quantity falls, per checkpoint. The transport boundary and the read-out band are fractions of depth, not fixed layer indices, which is what lets both policies transfer to a 36-layer model unchanged. The 36-layer boundary is masked: its suffix is harmless at L20 and zero from L24, the same fractions as the pair.

checkpoint	layers	block-mean band	read-out band	boundary	boundary / depth
Qwen3-VL-2B	28	L16–L26	L17–L20	L16	0.57
Qwen2-VL-7B	28	L15–L26	L19–L22	L16	0.57
Qwen2.5-VL-7B	28	L16–L26	—	L16	0.57
Qwen3-VL-8B	36	L20–L33	—	L21	0.57

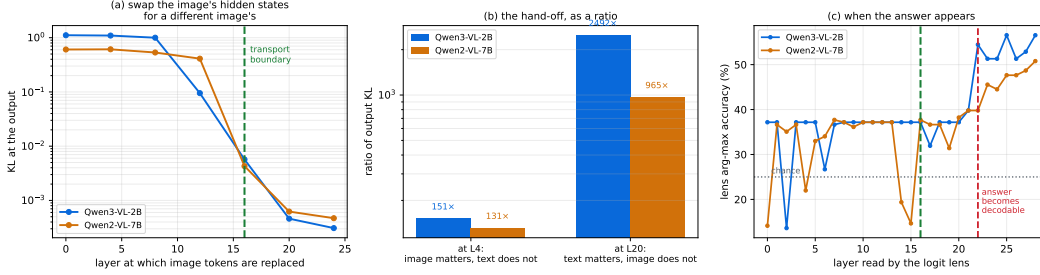


Figure 9: **(a)** Replacing the image tokens’ hidden states at layer ℓ with a different image’s: the output effect falls three orders of magnitude across the boundary. **(b)** The hand-off as a ratio: early, the image dominates and the text is inert; late, exactly the reverse. **(c)** Logit lens at the answer position: the correct option becomes decodable only in the last quarter of the stack.

E DEPTH STRUCTURE ACROSS CHECKPOINTS

F MECHANISM: THREE STAGES, THREE INTERVENTIONS

Each row of Table 19 is measured by a different intervention, so no stage rests on the method that found another.

Table 19: The three stages. Every number is measured on both core checkpoints.

stage	layers	measured by	evidence
image → text transport	L0–L16	attention masking; hidden-state patching	suffix masks cost nothing from L16 (0.014, 0.029); the patch effect falls 17× and 96× between L12 and L16
question-conditioned read-out	L17–L21	end-task accuracy by read-out depth	crop accuracy spans 34.6–70.2, peaking at L17; no layer inside the window beats not cropping
answer formation	L22+	logit lens	lens accuracy steps +18.0 [+8.7, +27.0] and +14.2 [+7.2, +21.3] above its early plateau

The logit lens is computed with the final norm applied to intermediate layers only; applying it to the last hidden state as well double-normalises and corrupts that layer alone, which cost us a retracted result earlier in this project. We assert per run that our last-layer read-out matches the true logits.

A fourth, independent technique places the read-out band in the same spot: a ridge probe from the answer-position state to the target’s location rises across the transport window and peaks at L18–19 (Qwen3, $r = +0.635$) and L18 (Qwen2, +0.603).

The channel and the values. Masking blocks the text→image attention *channel*; it does not show that the image information is absent from the residual stream. Replacing the image tokens’ hidden states with a donor image’s (same question, same token budget) closes that gap. The effect falls

Table 20: Representation patching: replace the image tokens’ hidden states at one layer’s output with a donor image’s; KL at the output and answer flips. Text controls patch text-token states at the same depth.

patch site	Qwen3-VL-2B KL (flips)	Qwen2-VL-7B KL (flips)
image L0	1.758 (61.7%)	0.800 (60.0%)
image L4 (control C1)	1.766 (63.3%)	0.798 (58.3%)
image L8	1.622 (60.0%)	0.657 (60.0%)
image L12	0.134 (16.7%)	0.517 (51.7%)
image L16	0.009 (3.3%)	0.005 (5.0%)
image L20	0.0005 (0%)	0.0008 (1.7%)
image L24	0.0003 (0%)	0.0006 (1.7%)
text L4	0.009 (5.0%)	0.006 (3.3%)
text L20 (control C2)	1.834 (63.3%)	0.802 (60.0%)

Table 21: Value inertness at 900 tokens (Qwen3-VL-2B, $n = 191$): donor hidden states at the layer’s output.

patch at 900 tokens	mean KL	answer flips
dropped positions @L16 (810 of 900)	0.0022	1.0%
kept positions @L16 (90 of 900)	0.0065	3.1%
dropped positions @L8 (transport window)	0.3407	20.9%

three orders of magnitude across the boundary, and both pre-registered controls behave as required: patching the image at L4 is large, patching text at L20 is large.

The late-image arm lands exactly on the text-L4 noise floor (0.009 vs 0.009 on Qwen3; 0.005 vs 0.006 on Qwen2), and the two controls invert cleanly across depth: at L4 the image matters $151 \times / 131 \times$ more than the text, at L20 the text matters $2492 \times / 965 \times$ more than the image. Two different interventions, two checkpoints, one boundary.

The boundary at the AVR operating point. AVR encodes at 900 tokens and prunes at the boundary, so inertness must hold on the unpruned 900-token pass. Injecting donor values at the layer’s output:

At the boundary the values are inert whether or not they are the ones AVR keeps; inside the transport window the same intervention moves one answer in five.

The cost of misplacing the boundary. Pruning is free at the boundary and expensive before it. With the same number of tokens dropped and a large negative attention bias on the dropped columns from layer K onward (functionally FastV’s rule):

No pruning is 56.5%. Ranking at layer 2 costs 22.0 points against no pruning and is 3.7 points *worse than random*; the late read-out deletes 90% of visual tokens at -0.5 points $[-5.8, +4.7]$ and the learned criterion at $+2.1$ $[-2.6, +6.8]$. The collapse replicates on Qwen2-VL-7B (late over layer-2 $+11.5$ $[+4.2, +18.8]$ at 10% keep; layer-2 is 4.7 points below random) and is larger on general VQA: layer-2 against random is -15.0 $[-22.0, -7.5]$ on POPE and -25.5 $[-33.0, -18.0]$ on MMBench.

Selection stops mattering at the boundary. Before the boundary, which tokens survive matters: a layer-4 probe beats random keep by $+8.4$ $[+2.1, +15.2]$ and $+8.9$ $[+2.6, +15.2]$ at 25% keep. At and after it, the choice is irrelevant: random keep matches attention keep on both models at every keep ratio tested, and pruning is free (0.0–0.5 points in 9 of 10 cells). The boundary is not merely where pruning becomes cheap; it is where selection ceases to matter.

Table 22: Pruning depth and selection. V*Bench, $n = 191$; every arm prunes the same count.

keep	random	layer-2 (FastV default)	late read-out (L16–26)
10%	38.2%	34.6%	56.0%
25%	41.4%	40.3%	58.1%
50%	47.1%	52.9%	55.5%

Table 23: The hand-off. Donor patching at L4 and L20, both checkpoints; ratios compare the two patch sites. Paired KL difference between the L16 and L4 image patches is $-1.757 [-2.308, -1.256]$ and $-0.792 [-1.031, -0.577]$.

site	image patch (Q3/Q2)	text patch (Q3/Q2)	ratio
L4	1.766 / 0.798	0.009 / 0.006	image $151 \times / 131 \times$
L20	0.0005 / 0.0008	1.834 / 0.802	text $2492 \times / 965 \times$

G LATE-DEPTH BEHAVIOUR: AGNOSTICISM, MERGING, AND DISAGREEMENT

The late layers are agnostic to the image. Three independent probes agree that direct image access stops mattering at the boundary: the suffix mask, the image-value patch, and the answer-row-only mask. The suffix mask is the same intervention as the boundary measurement; the value patch replaces the image tokens’ hidden states with a donor image’s; the answer-row mask blocks only the final prompt position’s image access at every layer.

Image and text tokens merge in the late stack. The patching controls invert across depth: early, the image values dominate and the text values are inert; late, exactly the reverse. Replacing image values past the boundary lands on the irrelevant-text noise floor, so the image tokens behave like ordinary, interchangeable context tokens whose information has already been absorbed into the text stream. The answer position confirms the direction of the hand-off: it cannot read the image directly, and the evidence arrives through the text representations that consolidated it.

The layers disagree about where to look. The 28 layer maps are not variations on one map. Mean off-diagonal arg-max agreement across the 28×28 layer pairs is 13.8%/38.1%, and the agreement matrix has block structure that splits at the transport boundary. The usable placement signal exists only after it.

Consequently, every label-free layer-disagreement rule from the language-model literature was ported to the read-out and compared with the fixed gate; none beats it, and most lose badly.

H FULL PAIRED COMPARISON AGAINST PUBLISHED RULES

Taking these six together with the three pooled block-mean contrasts in Table 26, DWA beats a published rule in nine of nine pooled comparisons. It does not beat the equal-compute bar everywhere, which is a separate claim and the subject of the scope law. The fold-seed control matters most. Refitting DWA under three disjoint fold seeds moves the pooled result by 1.0 point or less on both models, so the margin is not an artefact of which items landed in which fold. The oracle row is the remaining headroom: even the best rule we have is 15–17 points below perfect placement.

The pooled rows against the bar are the scope law in its plainest form. Cropping wins 9–10 points where the question is about one instance, loses 6–9 where it is about a relation between two, and the two cancel to nothing in aggregate. A paper that reported only the pooled column would conclude the method does not work; a paper that reported only the single-instance column would conclude it works everywhere. Neither is true.

Table 24: Label-free layer-selection rules, coverage@0.25 minus the fixed gate, four architectures.

rule	Qwen3	Qwen2	LLaVA-NeXT	OneVision
gate max (reference, coverage %)	56.0	51.8	25.7	23.6
CLD entropy-valley layer	-12.0	-1.0	+0.5	+4.2
attention-entropy-min layer	-2.6	+2.1	+0.0	+1.0
attention-JSD vs final	+1.0	-1.0	+1.6	-10.5
ASL rank-stability layer	-11.5	-8.4	+0.0	-6.8
depth-EMA of ranks	-13.6	-6.8	+0.5	+2.1
SLED extrapolation	+0.5	-0.5	+2.1	+2.1

Table 25: DWA minus each alternative on V*Bench, paired on identical items, pooled over strata, $n = 191$ (190 for LASER, one item missing). All six published-rule contrasts clear zero on both models. Bootstrap 95% CIs, 8,000 resamples, matching the protocol in Appendix A.

contrast	Qwen3-VL-2B	Qwen2-VL-7B
- uniform@600 (equal-compute bar)	+8.9 [+0.5, +17.3] ✓	+12.0 [+4.7, +19.4] ✓
- fixed-layer crop (L14)	+34.6 [+25.7, +43.5] ✓	+31.9 [+24.1, +39.8] ✓
- block-mean arg-max	+10.5 [+4.2, +16.8] ✓	+11.0 [+4.7, +17.8] ✓
- LASER [†]	+7.9 [+1.1, +14.7] ✓	+8.9 [+3.1, +14.7] ✓
<i>controls, not baselines</i>		
- gate-max	+3.7 [-2.1, +9.4]	+9.4 [+3.7, +15.2] ✓
- gradient-boosted tree	-0.5 [-4.7, +4.2]	+7.3 [+2.1, +13.1] ✓
- DWA refit, disjoint fold seeds	-1.0 [-3.1, +1.0]	+1.0 [-1.0, +3.1]
- oracle placement	-15.2 [-22.5, -8.4] ✗	-17.3 [-24.6, -9.9] ✗

I THE SIZE CLIFF

Protocol. Every measurement in this appendix uses V*Bench with its ground-truth boxes, on Qwen3-VL-2B and Qwen2-VL-7B unless stated. A *merged visual token* is one post-merge patch embedding; at a 300-token pass it covers $1/300$ of the image area. A target’s extent is $S \times B$ merged tokens, where S is the ground-truth box area as a fraction of the image and B the realised token count; its side is $\sqrt{S \times N}$ tokens. The *oracle crop* replaces the model’s window with the ground-truth box at the same 300-token budget and is a control, not a method. Thresholds come from a sliding sweep: for each t , accuracy below and at-or-above t is compared on single-region items, requiring at least ten items per side, and t^* maximises the step. All intervals are paired item bootstraps (8000 resamples).

The cliff. Accuracy as a function of the target’s extent, with the oracle crop as the difficulty control, is the central measurement of this appendix. The uniform curve is flat and near chance for the smallest targets, dips *below* the 25% chance level, and rises steeply through the shaded 0.15–0.25 merged-token band; the oracle curve is flat at 95.8%/100.0% across the same range. The two models’ curves differ in level but not in shape or in where the step sits. The step is sharpest at $t = 0.15$ on the cleanly-localised single-region subset, where the intervals are disjoint, and it appears at the same 0.25-token threshold on all single-region items of both models.

It is not the budget. The natural objection is that the cliff is simply a budget effect: spend more tokens and the target stops being sub-token. It is not. Recomputing the sliding threshold at 300 and 588 tokens shows that the threshold in merged tokens scales with the budget ($0.25 \rightarrow 0.55$ on Qwen3) while its image-side fraction is unchanged ($2.89\% \rightarrow 3.06\%$); Qwen2 is noisier but agrees in direction. In other words, the criterion is the target’s *relative size*, not the number of tokens that land on it: at a fixed tokens-on-target, accuracy differs between the two budgets (bin $[0.15, 0.25]$: 33.3% vs 53.8% on Qwen3; 41.7% vs 61.5% on Qwen2), so the two budgets do not collapse onto one curve. Uniform budget cannot substitute for magnification because it leaves the target’s fraction of the image unchanged.

Table 26: HR-Bench, DWA minus each alternative, $n = 400$ per stratum, 800 per cell. DWA is transferred: its weights were fit on V*Bench and applied here with nothing refit. Placement quality replicates; the scope law replicates with it.

		HR-4K Qwen3-2B	HR-4K Qwen2-7B	HR-8K Qwen3-2B
vs. bar	single	+10.2 [+5.5, +15.0] ✓	+9.2 [+4.5, +14.2] ✓	+8.8 [+3.5, +13.8] ✓
	cross	-8.8 [-14.0, -3.5] ✗	-6.0 [-11.5, -0.8] ✗	-8.8 [-14.5, -2.8] ✗
	pooled	+0.8 [-2.8, +4.4]	+1.6 [-1.9, +5.2]	+0.0 [-3.9, +4.0]
vs. block-mean	single	+7.8 [+4.0, +11.5] ✓	+15.0 [+10.2, +19.8] ✓	+5.2 [+1.2, +9.2] ✓
	cross	+1.8 [-2.8, +6.2]	+16.8 [+12.0, +21.5] ✓	+2.2 [-2.5, +6.8]
	pooled	+4.8 [+1.9, +7.6] ✓	+15.9 [+12.6, +19.2] ✓	+3.8 [+0.6, +6.9] ✓
vs. random cell	single	+32.8 [+26.5, +39.0] ✓	+24.8 [+18.8, +30.5] ✓	+24.2 [+18.0, +30.2] ✓
	cross	+10.8 [+5.0, +16.5] ✓	+15.8 [+10.5, +21.0] ✓	+7.0 [+2.0, +12.0] ✓
	pooled	+21.8 [+17.5, +26.1] ✓	+20.2 [+16.4, +24.2] ✓	+15.6 [+11.6, +19.6] ✓

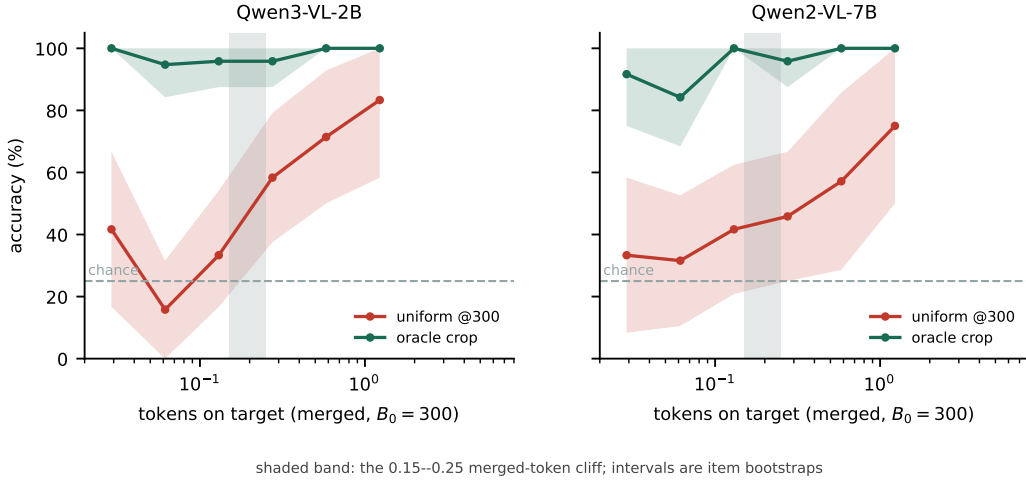


Figure 10: **The encoding cliff.** Accuracy vs target extent in merged tokens ($B_0=300$), with item-bootstrap intervals. Red: uniform encoding; green: oracle crop (same question, same budget, target resolved). Shaded band: the 0.15–0.25 merged-token cliff. The uniform curve passes *below* the chance line before the threshold and through it at the threshold; the oracle curve is flat.

Magnification is not the bottleneck. One control bounds the interpretation: the tightest oracle window tested, $W=0.15$, resolves 100% of items on both models in every size quartile (median $W^* = 0.15$; rank correlation with $\sqrt{S} \approx 0$). Once the region is isolated, even a $6.7\times$ crop reads the smallest detail, so what limits the model is isolating the region and the target’s relative size, not the amount of magnification.

J THE INFORMATION IS NEVER ENCODED

Protocol. An item is a *certified encoding failure* when it is wrong under the uniform bar and right under the oracle crop at the same budget; this is the label no other work can construct, because it needs the crop control. The layer-wise read-out is a logit lens: intermediate layers are decoded as $\text{lm_head}(\text{final_norm}(h_\ell))$, while the final layer is read from the model’s own logits, because applying the final norm to the last hidden state as well double-normalises that layer alone and corrupted an earlier result of ours (§F), we assert per run that the last-layer read-out matches the true logits. Representation patching replaces the image tokens’ hidden states at a single layer’s output with those of a donor image carrying the same question at the same token budget; the pre-registered controls are that patching the image at L4 must be large and patching text at L20 must be large.

Table 27: **The encoding cliff.** Single-region localised items ($n=63$, Qwen3-VL-2B); the oracle crop is flat across the same split (100.0% below, 97.6% above).

t (merged tokens)	n below	acc below	n above	acc above	step
0.10	16	18.8%	47	72.3%	+53.6
0.15	21	19.0%	42	78.6%	+59.5
0.25	33	33.3%	30	86.7%	+53.3
0.50	46	47.8%	17	88.2%	+40.4
1.00	57	56.1%	6	83.3%	+27.2

Table 28: **The cliff is architecture-invariant.** All single-region items, $n=115$ each; intervals are disjoint on both models, and the oracle crop is flat and identical.

model	cliff	uniform below	uniform above	oracle below	oracle above
Qwen3-VL-2B	0.25 tok	32.4%	77.3%	95.8%	100.0%
Qwen2-VL-7B	0.25 tok	36.6%	63.6%	95.8%	100.0%

The answer is not in the stack. If the answer were present but mis-routed, a layer-wise read-out would find it somewhere and lose it later. Under the uniform encoding no layer’s decoded answer exceeds 52.9%, and localisation quality does not change that: on items where the window covers the target and the model still answers wrong ($n=36$), the correct option is the argmax at some layer on 72.2% of items, against 68.1% where the window missed entirely. There is no layer at which the answer is present and subsequently lost: the apparent signal at L20 (41.7%) collapses to 0% from L22 and matches what near-chance layers produce by chance. Under the oracle crop the lens instead reaches 92.6%, and the answer appears abruptly at L22, taking 90% of the gain in a single transition; the step replicates one layer later on Qwen2-VL-7B and appears only where the crop covers the target (+23.8 points covered, exactly 0.0 where it misses). Cropping does not make the model reason better; it supplies information that a fixed depth converts into an answer. Representation patching agrees from the other side: replacing every image token’s hidden state with a different image’s, at a single layer, changes the answer no more than replacing irrelevant early text once past the transport boundary (KL 0.009 vs 0.009 on Qwen3 and 0.005 vs 0.006 on Qwen2 at L16). The oracle crop fixes 94.4% of the localised-but-wrong items at the same budget, with a median of 0.14 merged tokens on target, and across the benchmark 37.7%/41.4% of all items are certified encoding failures: 86.7%/84.0% of every error, rising to 94.0%/89.4% at $W=0.15$.

Why the target is invisible. The arithmetic behind the threshold is simple. At the 300-token pass the median single-region target’s side is 0.210 merged tokens; 100% of targets are smaller than one merged token, the median target occupies 0.176 tokens of area, and 62% sit below the 0.25-token cliff. The evidence is averaged into a single token’s footprint together with its background before the language model ever sees it — which is why magnification, not budget, is the operation that recovers it.

Controls. Two further controls bound the interpretation. *Localisation degrades below the cliff, far less than accuracy:* ground-truth coverage falls from 0.637 to 0.457 (argmax window) and from 0.751 to 0.580 (head window) on Qwen3, and $0.594 \rightarrow 0.450 / 0.620 \rightarrow 0.471$ on Qwen2, while accuracy falls by 42.6 points — encoding dominates, placement contributes a minority share. *The below-chance failure is not the language prior:* a text-only control (question without the image) agrees with the image-conditioned answer on 49.4% of below-cliff items vs 40.7% above on Qwen3, but 49.4% vs 56.5% on Qwen2 — opposite directions, neither interval clear. Below the cliff the model is systematically misled, but not by the text-only prior.

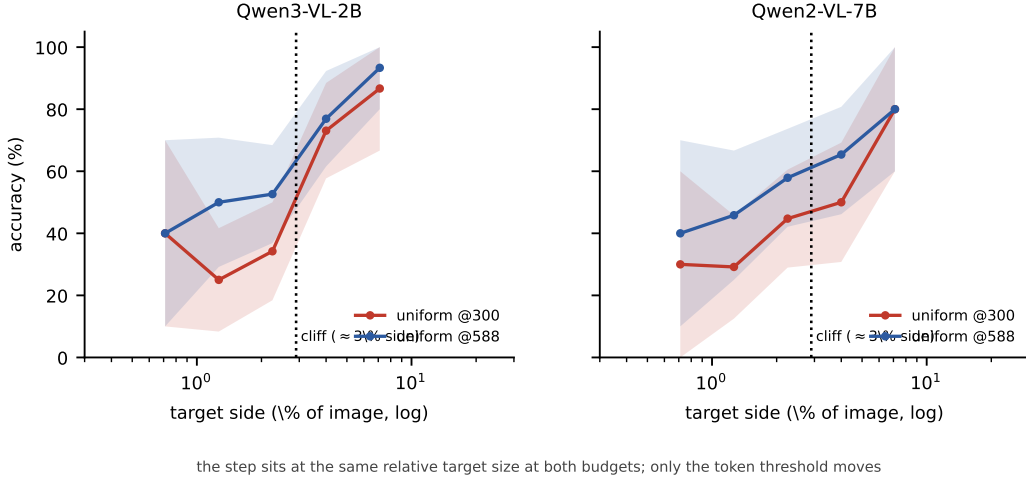


Figure 11: **The cliff does not move with budget.** Accuracy vs the target’s side as a fraction of the image, at 300 and 588 tokens, both models. The step sits at the same relative size ($\approx 3\%$ of the side) at both budgets: doubling the uniform budget moves the token threshold but not the cliff.

Table 29: **The cliff does not move with budget.** Single-region items; t^* maximises the below/above step.

model	budget	t^* (tok)	acc below	acc above	t^*/B (area)	side fraction
Qwen3-VL-2B	300	0.250	32.4%	75.0%	0.083%	2.89%
Qwen3-VL-2B	588	0.550	50.0%	87.2%	0.094%	3.06%
Qwen2-VL-7B	300	0.850	40.4%	75.0%	0.283%	5.32%
Qwen2-VL-7B	588	0.750	50.0%	77.4%	0.128%	3.57%

K ALLOCATION BEATS BUDGET

Protocol. The *exchange rate* compares an oracle crop at the base budget with the best uniform rung each model can reach; a random crop of the same size bounds the placement effect from below. No uniform rung reaches the crop arm on any model, so every rate is a lower bound set by the model’s token ceiling; LLaVA-NeXT has no resolution axis and is reported as such. The native-resolution comparison runs the same items with the uniform arm free to spend the model’s full dynamic range, and the boundary test runs HR-Bench 4K, where targets are large enough to be resolvable at the base budget.

Allocation vs budget. The exchange rate is the ratio of the token counts at which the two arms score what they score. An oracle crop at 300 tokens reaches 97.4% on Qwen3-VL-2B and 93.2% on Qwen2-VL-7B, while no uniform allocation on any rung up to 7,776 tokens reaches the crop arm (Qwen2’s ceiling is 75.4%); the implied exchange rate is $\geq 26\times$ and $\geq 25.9\times$ the same tokens spent uniformly, with the placement effect itself worth +40.0 and +59.2 points. At native resolution the same holds: a perfectly placed 600-token crop beats native dynamic resolution (3,290 tokens, $5.5\times$ the budget) by +11.6/+16.2 pooled. The boundary is measured rather than assumed: allocation pays when the evidence is smaller than the budget can resolve — 71% of V*Bench targets are sub-token and oracle allocation is worth +37.2 points — and costs when it is not, as on HR-Bench 4K, where allocation loses 8.8 points because discarded context dominates. The localizer is good in both regimes; what changes is whether acting on it helps.

What did not extend the crop. Two attempts to buy the crop out of its cross-instance limit failed, and both are charged in full. Adding the original image back costs $1.5\times$ the bar once the localisation pass is counted and scores below the bar-matched crop alone on every stratum (pooled +4.2 vs

Table 30: **No collapse: accuracy at fixed tokens-on-target, by budget.** Qwen3-VL-2B / Qwen2-VL-7B.

tokens-on-target	acc @300 (Q3/Q2)	acc @588 (Q3/Q2)
[0, 0.15)	31.9 / 31.9	48.5 / 39.4
[0.15, 0.25)	33.3 / 41.7	53.8 / 61.5
[0.25, 0.50)	73.7 / 63.2	52.0 / 60.0
[0.50, 1.00)	80.0 / 60.0	80.0 / 65.0
[1.00, 2.00)	66.7 / 50.0	78.6 / 78.6

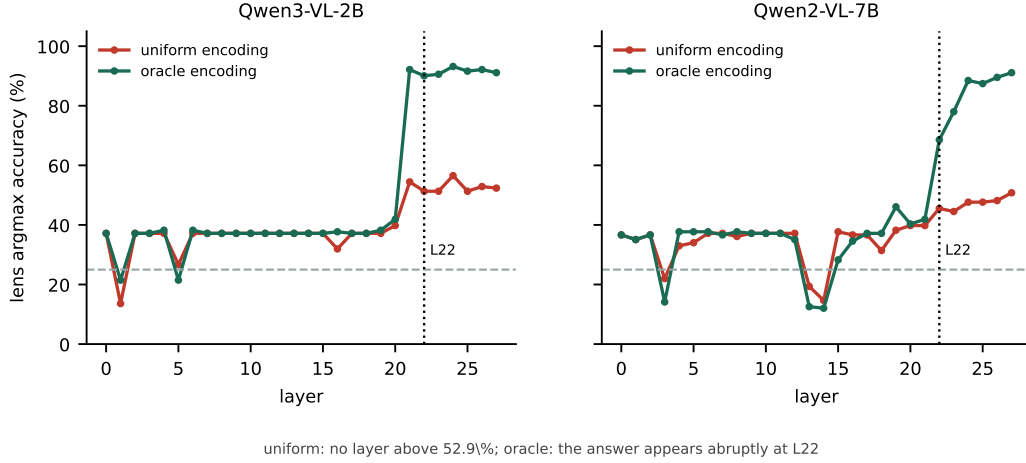


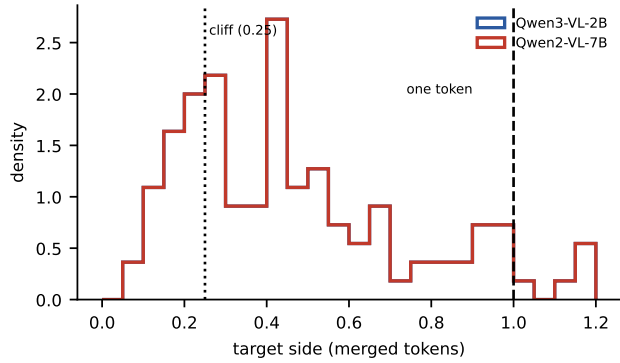
Figure 12: **The answer is not in the stack.** Per-layer logit-lens accuracy under the uniform encoding (red) and the oracle crop (green), both models. Uniform: flat, no layer above 52.9% and none holding an answer that is later lost. Oracle: the answer becomes decodable abruptly at L22, taking most of the gain in one transition; the Qwen2 step is one layer later.

+10.5; cross -10.5 vs -1.3); the original dilutes the crop’s attention more than it restores the second object. Two full-resolution crops, funded by early-exit localisation and boundary pruning at 101–104% of the bar, help single further (80.9 vs 62.6) but do not clear on cross (68.4 vs 65.8, 0 of 2 pre-registered).

L THE LITERATURE PROTOCOL, REPRODUCED

The cropping literature compares a crop against no crop at the *same* answer budget and does not charge the localisation pass. To read our results against those tables, we reproduced those rules exactly on our checkpoints: no-crop is the model on the full image, the crop is placed by the ridge at the same token count as the no-crop arm, and the localise pass is not charged. Only benchmarks that overlap the prior work are shown.

Readings. (i) The family’s V*Bench result reproduces on our crops and our models: +13.6 to +21.7, against the published +19.9 (LLaVA-1.5) and +6.8 (InstructBLIP). (ii) The published DocVQA result reproduces exactly where the baseline is resolution-starved: LLaVA-OneVision at its ladder minimum (no-crop 32.7) gains +9.4 from cropping, in the same direction as the published +1.6 to +3.9 against a 336px baseline, while Qwen3-VL-2B at the same 299-vs-299 tokens (no-crop 63.9) loses 11.2. (iii) TextVQA is the controlled case discussed in §5.4: LLaVA-1.5 at fixed 336px gains, LLaVA-OneVision anyres at 1,286 tokens loses. (iv) GQA is negative for an anyres model, where the published result is flat — consistent with the scope law, since GQA is relation-heavy.



every single-region target is smaller than one merged token; 62% sit below the cliff

Figure 14: **The target is smaller than a token.** Distribution of the single-region target’s side in merged-token units at the 300-token pass, both models. Every target lies left of the one-token line; most lie below the 0.25-token cliff.

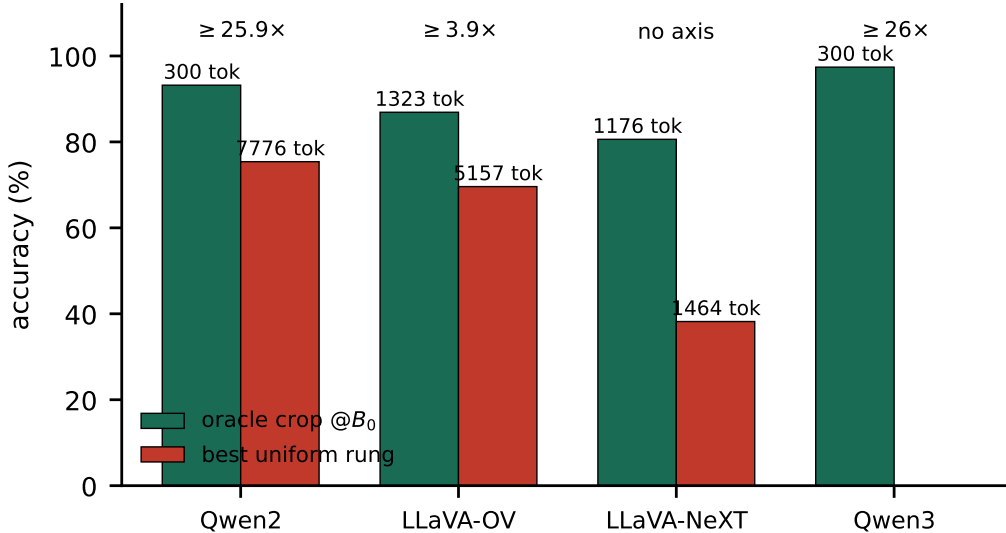


Figure 15: **Allocation vs budget.** Oracle crop at the base budget (green) against the best uniform rung each model reaches (red; token counts annotated). The exchange rate is the ratio of the token counts at which the two arms score what they score; no uniform rung matches the crop, so each rate is a lower bound.

M THE RESOLUTION-HEADROOM IDENTITY

The identity also holds where it predicts *nothing*. Applying AVR to a crop rather than to the full image gives eight further cells whose headroom is near zero; gain tracks headroom there with $r = 0.911$ and slope 1.03 (mean error 0.40 points), and AVR correctly does nothing. Across both settings the identity is supported on 18 stratum-cells, and it is the only component of this work that predicted a negative result before we ran it.

Two consequences. First, the sign of the gain is predictable before deployment: the one cell where AVR loses (HR-4K cross-instance) is the one cell where the baseline itself is worse at 900 tokens than at 600, so nothing was available to convert. Second, the pruning contributes nothing on its own. Comparing AVR against `uniform@900`, which encodes at the same resolution and prunes

Table 32: **Extensions of the crop, all charged in full.** Qwen3-VL-2B, V*Bench, $n=191$; deltas against the equal-compute bar. The funded 2-crop’s pre-registered criterion did not pass (0 of 2).

arm	cost vs bar	pooled	single	cross
DWA crop (reference)	1.0×	+10.5	+18.3	−1.3
crop + original	1.5×	+4.2	+13.9	−10.5
2 crops + original	1.5×	+7.3	+13.9	−2.6
4 crops + original	1.5×	+5.8	+11.3	−2.6
funded 2-crop	1.01–1.04×	—	+18.3	+2.6

Table 33: **Our checkpoints under the literature’s own protocol** (equal answer tokens, localise pass uncharged). Crop = the better of $W=0.25$ and $W=0.5$. Deltas are paired bootstrap 95% CIs; ✓ = CI excludes zero. †partial cell.

model	benchmark	n	no-crop	crop	Δ
Qwen3-VL-2B	V*Bench	191	53.9	72.8	+18.8 [+11.0, +26.7] ✓
Qwen3-VL-2B	TextVQA	500	76.8	70.8	−6.0 [−10.2, −1.8]
Qwen3-VL-2B	DocVQA	527	63.9	52.8	−11.2 [−15.9, −6.5]
Qwen2-VL-7B	V*Bench	120†	48.3	70.0	+21.7 [+12.5, +30.8] ✓
LLaVA-OV-7B	V*Bench	191	51.8	65.4	+13.6 [+6.3, +20.9] ✓
LLaVA-OV-7B	TextVQA	5000	67.9	63.8	−4.2 [−5.7, −2.6]
LLaVA-OV-7B	DocVQA	1000	32.7	42.1	+9.4 [+5.8, +13.0] ✓
LLaVA-OV-7B	GQA	248†	61.3	51.6	−9.7 [−15.3, −4.0]

nothing, the two agree within 0.5 points in 7 of the 10 cells (+0.0, +0.0, +0.0, +0.0, +0.2, +0.5, −0.2). In two more the pruned arm is *ahead*, by +1.3 and +2.6, which we read as noise rather than a benefit of pruning. Exactly one cell shows a real cost: HR-8K cross-instance at −1.8 [−3.2, −0.5], the only measurable price of discarding 90% of visual tokens anywhere in this work. Pruning at the boundary is therefore free everywhere we have measured it but that one cell. We verified the pruning is applied rather than silently skipped at the distribution level: output probabilities change on 99% of items while the arg-max changes on 2%. The discarded tokens are inert, not absent.

N CLAIM-TO-EVIDENCE MAP

Every headline claim, its strongest evidence, and its model count. *Intervention* means the claim was tested by changing the model’s computation (mask, patch, prune, ablate, oracle), not by observation alone. A claim is exported only if it holds on two checkpoints.

O SCOPE AND COMPUTE, AS FIGURES

P PROPOSITIONS, AND WHAT VALIDATES THEM

We state four theorems, three propositions and two lemmas. They are not deep results — two are tail-dominance and sufficiency statements, two are information bounds, and the lemmas are one-line linear algebra — but each makes a prediction that could have failed on data, and we report where they hold approximately rather than exactly. All measurements are out-of-fold on both core checkpoints; $n = 126$ (modal grid) for coverage, $n = 191$ for end-task and distributional tests.

Setup. For item i , layer ℓ , cell c , let $A_\ell^{(i)}(c) \geq 0$ be the last-prompt-token attention mass on cell c , row-normalised so $\sum_c A_\ell^{(i)}(c) = 1$. Write $m_\ell = \mathbb{E}_i A_\ell^{(i)}$ for the *item-mean* (item-independent) map and $A_\ell^{(i)} = m_\ell + U_\ell^{(i)}$, so $\mathbb{E}_i U_\ell^{(i)} = 0$.

Theorem 1 (Tail dominance of the conventional read-out). *Write each depth’s per-cell score as $x_\ell(c) = m_\ell(c) + u_\ell(c)$ with u_ℓ zero-mean over items, and put $\Delta_B = \max_c \sum_{\ell \in B} m_\ell(c) - \text{med}_c \sum_{\ell \in B} m_\ell(c)$. The block read-out returns $\arg \max_c \sum_{\ell \in B} x_\ell(c)$.*

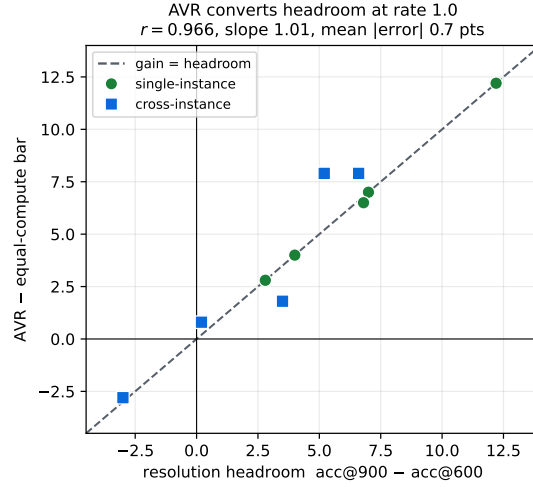


Figure 16: **AVR is a resolution converter.** Its gain over the equal-compute bar tracks the headroom the checkpoint already has, at rate 1.0. The one cell where AVR loses is the one cell where the baseline itself is worse at 900 tokens than at 600.

Table 34: AVR on HR-Bench. Gain over the equal-compute reference, the measured resolution headroom ($\text{acc}_{900} - \text{acc}_{600}$), and the pruning cost ($\text{AVR} - \text{uniform@900}$).

cell	stratum	AVR – reference	headroom	pruning cost
HR-4K Qwen3-VL-2B	single	+4.0 [+0.5, +7.5]	+4.0	+0.0
HR-4K Qwen3-VL-2B	cross	-2.8	-3.0	+0.2
HR-4K Qwen2-VL-7B	single	+2.8	+2.8	+0.0
HR-4K Qwen2-VL-7B	cross	+0.8	+0.2	+0.5
HR-8K Qwen3-VL-2B	single	+6.5 [+3.0, +10.2]	+6.8	-0.2
HR-8K Qwen3-VL-2B	cross	+1.8	+3.5	-1.8 [-3.2, -0.5]

- (i) If Δ_B exceeds the range of $\sum_{\ell \in B} u_\ell$, the arg-max is $\arg \max_c \sum_B m_\ell$ for every item: a fixed cell independent of the image.
- (ii) Equal-weight positive averaging cannot escape (i) at any band width: the m_ℓ leads are sign-aligned across depth, so Δ_B is superadditive in B .
- (iii) Any per-cell rescaling that equalises m across cells — in particular dividing by an estimate of m — removes the dominance without using labels.

Proof. (i) If the deterministic part’s lead exceeds the total variation the item-specific part can supply, it decides the maximum. (ii) Each summand contributes its m_ℓ lead with the same sign, so Δ_B is superadditive; whether Δ_B/σ_B rises or falls depends on the covariance of the u_ℓ , and the claim is only that it does not reach the $O(1)$ regime in which $\sum_B u_\ell$ could decide the arg-max. (iii) Immediate from (i) applied to the rescaled scores. $\square$

Lemma 1 (Leakage decomposition). *For every weight vector w , $\sum_\ell w_\ell A_\ell^{(i)} = L(w) + \sum_\ell w_\ell U_\ell^{(i)}$ with $L(w) = \sum_\ell w_\ell m_\ell$ independent of i . The item-independent component of the score is exactly $L(w)$.*

Proof. Substitute the decomposition and rearrange. $L(w)$ carries no i -dependence, so its ranking of cells is identical for every item. $\square$

Lemma 2 (Non-negative filters cannot cancel a common prior). *If the item-mean maps are rank-one across depth, $m_\ell = \alpha_\ell \mu$ with $\mu \neq 0$ and $\alpha_\ell > 0$, then $L(w) = (\alpha^\top w) \mu$. Hence $L(w) = 0$ iff $\alpha^\top w = 0$; no non-zero $w \geq 0$ achieves zero leakage.*

Table 35: Claims, evidence class, and the models on which the evidence was measured.

claim	strongest evidence	class	models
transport complete by $0.57 \times$ depth	layer-wise and prefix/suffix attention masks; 36-layer boundary from pruning cost	intervention	2×28 -layer masking; 36-layer pruning
image-token values inert past the boundary	representation patching at $L \geq 16$, at 300 and 900 tokens	intervention	2 models, both strata
answer position does not read the image	answer-row-only mask, all layers	intervention	2 models
depth choice dominates usable read-out band lies after transport	crop at every layer's arg-max, end task coverage curve, end-task sweep, location probe	intervention measurement intervention	2 models + 2 models (curve, probe)
ranking matters before the boundary only	probe vs random keep at L4; random = attention at L16/L24; value patching	intervention	2 models
block mean fails on the item-independent component	nuisance ablation: $0.083 \rightarrow 0.438$, $0.292 \rightarrow 0.490$	intervention	2 models
supervised depth weighting, not signed subtraction	non-negative fit retains the advantage (44/43 of 56 coefficients zeroed); end task n.s.	intervention	2 models
DWA beats every published rule at equal compute	9/9 pooled paired contrasts	evaluation	2 models $\times$ 2 benchmarks
AVR pruning is free	pruned vs unpruned @900; value patching at L16	intervention	2 models
AVR's gain is resolution headroom	10-cell identity, $r = 0.966$; distributional equivalence	accounting + intervention	4 checkpoints
cropping helps single-instance, not cross	14-cell scope table + oracle ceiling	evaluation + intervention	4 checkpoints
a map-only gate removes the cross loss	8-cell policy table: AND of three label-free statistics; HR-4K cross $+0.8/+1.0$, V^* Bench both strata clear	evaluation	2 models $\times$ 4 benchmarks
DWA matches native resolution at 14–18% compute	end-task vs native	evaluation	2 models
placement is worth $> 5 \times$ the compute	oracle-crop arm beats native	intervention	2 models

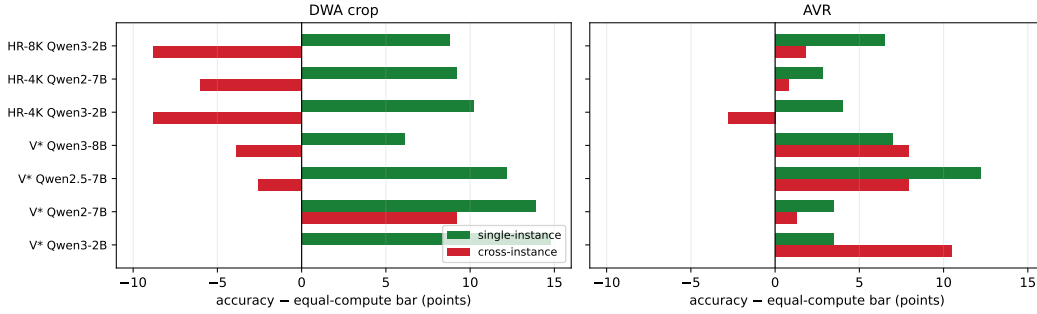


Figure 17: **The scope law.** Cropping is worth +6 to +15 points on single-instance questions in all seven cells and is negative on cross-instance in six of them. AVR is positive on both strata in six of seven, and gives up most of the single-instance gain to get there. The two policies are not variants of one method.

Proof. Linearity, then $\alpha^\top w = \sum_\ell \alpha_\ell w_\ell > 0$ whenever $w \geq 0$, $w \neq 0$. $\square$

Every part is measured, and the measurements decide between accounts that (ii) alone cannot separate. On the items the block mean gets wrong, the cell it selects sits at the **100th percentile** of the item-independent map — on both checkpoints, on every failure. It is not mis-weighting evidence; it is returning the raster sink. Consistent with (ii), Δ_B/σ_B measured over band widths 1 to 11 stays between 10.2 and 16.6 on one checkpoint and between 7.0 and 15.1 on the other — always an order of magnitude above the regime where the item-specific term could win, so no amount of averaging rescues the rule. The ratio *rises* with width on the first checkpoint and *falls* on the second, so we claim only that averaging never escapes, not that wider is monotonically worse. What is common to both is that the deployed eleven-wide band is worse than the best narrower band ($0.418 \rightarrow 0.083$ and $0.390 \rightarrow 0.297$), and that its last-column rate is higher there ($17.5 \rightarrow 91.3\%$ and $45.2 \rightarrow 62.7\%$). Consistent with (iii), an *unsupervised* correction — divide each cell by its value on a typical item, then take the arg-max — lifts coverage from 0.083 to 0.408 and from 0.292 to 0.504, reaching 71% and 86% of what the fitted read-out achieves, and drops the last-column rate to 2.4% and 0.0% against a 4.8% chance rate, with no ring mask.

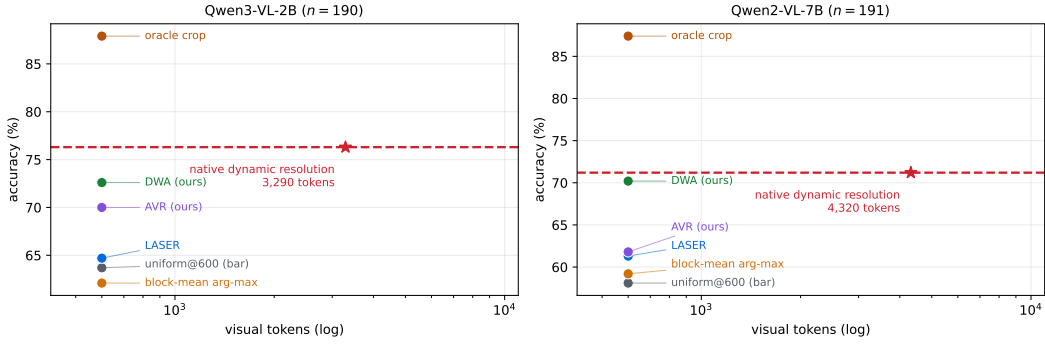


Figure 18: **Accuracy against visual-token compute, V*Bench.** The dashed line is the model run at its own native dynamic resolution. Our two policies sit on it at 14–18% of its compute; the published rules sit significantly below it; and a perfectly placed 600-token crop sits well above it. Placement is worth more than $5\text{--}7\times$ the tokens.

And it buys nothing at the end task, which bounds what this proposition explains. Run as a placement rule and answered, the same label-free correction scores 59.7% and 61.3% against the block mean’s 62.3% and 59.2%: $-2.6 [-7.9, +2.6]$ and $+2.1 [-3.1, +7.3]$, neither clear of zero, and it does not beat not cropping at all. DWA beats it by $+13.1 [+5.8, +20.4]$ and $+8.9 [+2.6, +15.2]$, clear on both checkpoints. So the proposition explains why the conventional read-out *places* badly, and a label-free fix repairs the placement, but repairing placement is not what the fitted read-out is selling. Its end-task advantage survives after the placement failure has been removed by other means, and we do not yet have a mechanism for that remainder. This is also the largest coverage-to-accuracy gap we have measured — 66–70% of the coverage gap closed, none of the accuracy gap — and it is a caution that applies to our own use of coverage in this section, not only to prior work.

What the fit adds, and why its sign does not matter. Because the features are log-attention, a weight vector summing to zero computes a *ratio* across depths, which cancels any per-cell factor common to all layers; the fitted weights are close to that, with $|\sum_{\ell} w_{\ell}| / \sum_{\ell} |w_{\ell}| = 0.126$ and 0.225 . But that is not the only route. Sink dominance itself varies $3.2\times$ and $4.4\times$ across depth, and the least-dominated depths are exactly the read-out band, so a strictly non-negative read-out can escape by *selecting* those depths rather than by cancelling anything. Both routes are available, which is why constraining every weight to be non-negative costs nothing while zeroing 44 and 43 of 56 coefficients. The item-mean maps are approximately rank-one (leading eigenvalue share 0.86 and 0.77), and in map space the non-negative constraint costs only 0.044 and 0.002 mean coverage: the sign of the fitted weights is an artefact of ridge collinearity between adjacent layers, not the mechanism. We do not claim the fitted weights track dominance depth-by-depth: that correlation is only $+0.171$ and $+0.131$. The supported statement is that the failure is tail dominance, that it has an unsupervised fix which recovers most of the gap, and that supervision buys the remaining 14–29%.

A refinement the theory predicted, which failed. If what remains after dividing out m were heteroscedastic across cells, the right label-free statistic would be the studentised one, $(x(c) - m(c))/s(c)$. We pre-registered that it would beat the plain ratio. It does not, on either checkpoint (-0.015 and -0.028), and it closes 63% and 61% of the gap against the plain ratio’s 66% and 70%. The proposition therefore explains the failure and supplies a label-free correction, but does not yet predict a better one: the residual that supervision buys is not simple heteroscedasticity.

Two accounts we tested first and rejected. That signed weights cancel a common nuisance: refuted, since non-negative fits lose nothing and the orthogonality the argument needs measures $+0.431$ and $+0.388$ rather than zero. That the fit is a matched filter over depth, weighting by per-layer signal-to-noise: refuted, one of two on each of its predictions, and wrong in kind — the block mean already retains 77% of attainable signal-to-noise on both models while its coverage is several times worse, and a mean-based quantity cannot explain an arg-max failure.

Theorem 2 (A single-depth read-out is lossy unless that depth is sufficient). *Let c^* be the cell containing the evidence and $A = (A_1, \dots, A_L)$ the depth profile. For any rule $\hat{c} = g(A_{\ell})$ reading a single layer: (i) $c^* \rightarrow A \rightarrow A_{\ell} \rightarrow \hat{c}$ is a Markov chain, so $I(c^*; \hat{c}) \leq I(c^*; A_{\ell}) \leq I(c^*; A)$; (ii) the*

right inequality is an equality iff $c^* \perp A \mid A_\ell$, i.e. A_ℓ is sufficient for c^* within the profile; (iii) by Fano, $\Pr[\hat{c} \neq c^*] \geq (H(c^*) - I(c^*; A_\ell) - 1)/\log |\mathcal{C}|$.

Proof. A_ℓ is a coordinate projection of A and $\hat{c}$ is post-processing of A_ℓ , so the chain holds and the data-processing inequality applies along each link, giving (i). For (ii), $I(c^*; A) - I(c^*; A_\ell) = I(c^*; A \mid A_\ell) \geq 0$ by the chain rule, with equality exactly under conditional independence. (iii) is Fano applied to that chain. $\square$

Theorem 2 is vacuous until sufficiency is decided, and sufficiency is an empirical property of a given model, so we test its equality condition. If A_ℓ were sufficient its arg-max would be correct wherever any layer’s is, so the best single depth would match the family’s envelope. Cropping at every one of the 28 layers’ arg-max and answering gives:

Table 36: The sufficiency test at the end task. A sufficient layer would make the first two columns equal.

	best depth	some depth	gap	every depth	spread over depth
Qwen3-VL-2B	70.2	95.3	+25.1	5.2	34.6–70.2
Qwen2-VL-7B	62.8	89.5	+26.7	9.9	36.1–62.8

Sufficiency is rejected on both checkpoints by 25 and 27 points. The escape is to read every depth: DWA is +2.6 [−3.7, +8.9] and +7.3 [+1.0, +13.1] against the best single depth — a depth chosen *on the test set*, so an optimistic baseline — clear on one checkpoint and matching on the other, while the published fixed choice sits 31.9 and 19.9 points below that oracle. The scope of the test is stated: the envelope is measured over the arg-max read-out family, which is the class a depth-selection rule chooses among, not over all measurable functions of A .

Theorem 3 (Decoding bound, and a certified gap for the conventional read-out). *For any read-out decision $\hat{c}$ of a per-cell score and target cell T : (i) $I(\hat{c}; T) \leq H(\hat{c}) \leq \log |\mathcal{C}|$; (ii) $I(\hat{c}; T) \geq H(T) - H_b(P_e) - P_e \log(|\mathcal{C}| - 1)$ with $P_e = \Pr[\hat{c} \neq T]$; (iii) if $\hat{c}$ is constant then $I(\hat{c}; T) = 0$.*

Proof. (i) $I(\hat{c}; T) = H(\hat{c}) - H(\hat{c} \mid T) \leq H(\hat{c}) \leq \log |\mathcal{C}|$. (ii) Fano’s inequality. (iii) $H(\hat{c}) = 0$, so (i) gives $I = 0$. $\square$

The bounds are exact and directly measurable, and the same maps admit an oracle decision that *is* the target, so $I(\text{oracle}; T) = H(T) = 6.06$ bits on the modal-grid items. The conventional read-out does not reach that: the literal block mean (no ring mask) has decision entropy 1.02 and 3.46 bits on the two checkpoints and lands in the grid’s last column on 84.8% and 61.8% of the full item set, so it can carry at most that much — a certified gap of ≥ 5.0 and ≥ 2.6 bits. The ring mask restores the baseline’s decision entropy (6.30/4.60 bits) but not its accuracy; the fitted read-out attains the same entropy with higher coverage (0.605/0.581 against 0.459/0.402). The failure is in the decoder, not the representation. Two consequences. The information a decision can carry is raised only by making it *varied* or *accurate* — the bounds never mention cancellation — so a non-negative depth fit reaches the same terms by selecting the depths where the item-independent component is least dominant (§3.4). And the depth *ordering* changes both terms while the sign of the weights does not.

Theorem 4 (The information horizon). *Suppose the visual-token values after layer p are output-invariant: $I(V_{>p}; Y \mid \text{state}_p) = 0$ (value-inertness). Then (i) pruning to any keep-set loses no answer information; (ii) AVR’s gain over the bar E_0 is the resolution information gain $\text{acc}(E) - \text{acc}(E_0)$, bounded by $I(X; Y) - I(Z_{E_0}; Y)$; and (iii) the information-bearing depth is $p + 1$ layers, so the affordable resolution multiplier is $N/(p + 1) = 1.647\times$ over uniform-depth encoding.*

Proof. (i) The premise makes the answer conditionally independent of those values given the state at p , and the keep-set is a function of them. (ii) Pruning is information-free by (i), so AVR and uniform encoding at E induce the same distribution; the data-processing inequality bounds the remaining difference by the information the bar’s encoding fails to carry. (iii) Token-layer accounting with the information-bearing depth. $\square$

The premise is measured with its violation bounded by Pinsker, $TV \leq \sqrt{KL/2}$: at the boundary the answer distribution depends on the token values by at most 0.033–0.044 in total variation, while the same patch one window earlier moves it by 0.36–0.41 (Appendix F). The horizon is sharp, and it is the same $0.57\times$ depth the masking experiments locate. The conclusion is what Appendix M tests: gain tracks headroom with $r = 0.966$ over ten cells, and again with $r = 0.911$ over eight further cells whose headroom is near zero, where it correctly predicts that AVR does nothing. *Where it is only approximate*: the premise holds to $KL \approx 5 \times 10^{-4}$, not to zero, so the conclusion is approximate too. Measured slippage is at most 0.5 points in seven of ten cells and -1.8 in one. The theorem’s real content is that the *choice* of keep-set is irrelevant, which is why random keep matches attention keep on 96.3% of items.

Proposition 1 (The budget determines the affordable resolution). Pruning to a fraction k at layer P of N costs $TL = E(P+1) + kE(N-1-P)$ token-layers. Matching a uniform bar of B tokens gives $E^* = BN/((P+1) + k(N-1-P))$. With $B = 600$, $N = 28$, $P = 16$, $k = 0.1$ this is $E^* = 928$, a $1.55\times$ resolution multiplier. We ran $E = 900$, 97% of the optimum, and realised 96–100% of the bar. This is arithmetic; we state it because it is the design equation, and because it says the multiplier is fixed by the boundary’s depth fraction rather than chosen.

Proposition 2 (Value-inertness implies selection irrelevance). Suppose that for every subset D of visual positions, replacing the hidden values of D at layer p with arbitrary donor values leaves the option distribution unchanged. Then for any two keep-sets D_1, D_2 of equal size, the models $U_E \circ P_{D_1}$ and $U_E \circ P_{D_2}$ induce the same distribution: every ranking rule is equivalent.

Proof. The premise makes the output invariant to the values of any subset of visual positions at layer p . The kept set is such a subset, so no choice of kept set can change the distribution. $\square$

Validation. On the unpruned 900-token pass at the boundary (L16, Qwen3, $n = 191$), replacing the hidden states of the 810 pruned positions with another image’s gives KL 0.0022 (1.0% flips) and replacing the 90 kept positions gives KL 0.0065 (3.1%); the same patch at L8 gives 0.341 (20.9%). Qwen2 replicates (0.0038/3.7% and 0.0027/0.5% at L16; 0.2622/29.8% at L8), so the premise holds on both models.

Proposition 3 (A single window cannot serve two distant sites). A window of side W contains at most one of two evidence sites separated by more than W . *Trivial, and it is not the interesting part. The interesting part is empirical: with perfect placement, cross-instance error stays at 25% and 22% against 3.5% and 6.1% for single-instance, and two full-resolution crops covering both objects still score -2.6 and -3.9 against one. The residual is not a perceptual failure, so no reallocation of perceptual resource reaches it — which is why the scope law is a property of the question type and not of the method.*

Q ADDITIONAL VISUALISATIONS

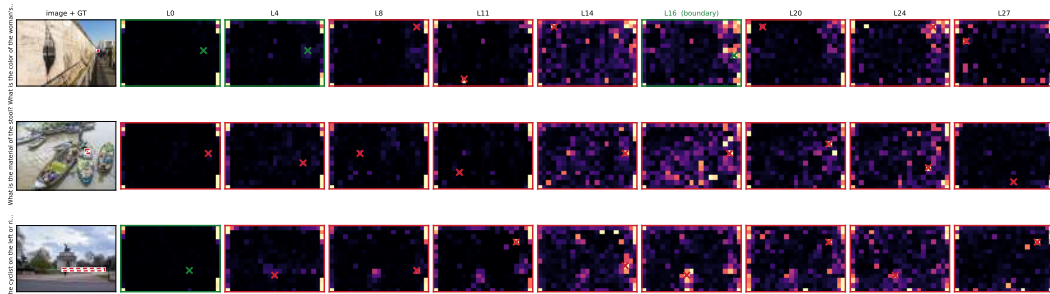


Figure 19: **The same image, the same question, a different answer about where to look at every depth.** Each panel is one layer’s attention over image cells (colour clipped at the 98th percentile so the serialisation sink does not wash out the structure); the cross is that layer’s ring-masked arg-max, green where the resulting crop would cover the evidence and red where it would not. Nothing about the input changes across a row. Only the read-out depth does.



Figure 20: **Left, centre: two layers rarely choose the same cell.** Mean off-diagonal agreement is 13.8% and 38.1%; the block structure splits at the transport boundary (dashed). **Right: and only the layers after the boundary are worth reading.** The fraction of items whose single-layer crop covers the evidence is at or below 20% everywhere before the boundary, jumps immediately after it, and decays again toward the output. This is the paper’s central claim in one curve: the usable placement signal exists only after transport has finished.



Figure 21: **Five placement rules, same items, same budget.** Selection rule: single-instance items on which the oracle crop is correct, so the ceiling behaves like a ceiling, and on which DWA is correct while the fixed-layer rule is not; 61 items qualify and the first four by question id are shown. The fixed mid-layer recipe crops glass, masonry and sky.

R METHOD ABLATIONS

Window size. W is a first-order hyperparameter, and the right value is set by the same condition that sets the sign. At an equal answer budget (300 vs 300 tokens, identical items), V*Bench wants the tight window: $W=0.25$ scores 72.8 against $W=0.5$ ’s 62.3 and the full image’s 53.9. On text and documents the ordering reverses — the tight window clips words and drops the referent — and $W=0.5$ recovers most of the gap: TextVQA 49.8 $\rightarrow$ 67.9 and DocVQA 35.2 $\rightarrow$ 50.3 at the same budget, still below the no-crop arms (74.7 and 56.0). A dynamic rule that sizes the window from the ridge score’s top-5% bounding box degenerates to a wide window (mean W 0.46–0.47; 64–71% of items at the 0.50 clamp) and tracks $W=0.5$ rather than adapting; reported as a negative for that rule, not for dynamism in general.



Figure 22: **Eight inferences, Qwen3-VL-2B.** Top of each cell: the image with the ground-truth box, the block-mean crop and the DWA crop. Below: the two crops the model actually answers from, with its answer.

Per-head statistics. The deployed read-out averages over heads. Adding, per layer, the per-cell maximum over heads and the head-distribution std (91 features instead of 63) raises out-of-fold coverage from 0.605 to 0.630 on Qwen3 and 0.581 to 0.609 on Qwen2. End-task the gain is +1.0 [−4.2, +6.3] on V*Bench and +2.4 [−1.0, +5.8] on TextVQA: positive in both cells, significant in neither. The coverage gain does not convert — another instance of the intermediate-metric gap (Appendix C) — so the head-augmented filter is kept as an option rather than as the deployed method.

S THE MAP-ONLY POLICY IN FULL

Policy — equal-compute bar, paired item bootstrap (8000 resamples). The gate is the AND of disp, band_disp and VRH above their item-set medians; the arms are DWA@300 (crop) and AVR (whole image). Strata are only defined on V*Bench and HR-4K. Script: phase237_policy_final.py.

Table 37: **Full intervals for the map-only policy.**

model	benchmark	route%	policy — bar	single	cross
Qwen3-VL-2B	V*Bench	36	+10.5 [+4.2, +17.3]	+9.6 [+0.9, +18.3]	+11.8 [+3.9, +19.7]
Qwen3-VL-2B	HR-4K	32	+3.9 [+1.4, +6.4]	+7.0 [+3.0, +11.0]	+0.8 [−2.3, +4.0]
Qwen3-VL-2B	CV-Bench	35	−3.3 [−4.6, −2.0]	—	—
Qwen3-VL-2B	RealworldQA	34	+0.8 [−1.8, +3.4]	—	—
Qwen2-VL-7B	V*Bench	38	+7.9 [+1.6, +14.1]	+10.4 [+2.6, +18.3]	+3.9 [−5.3, +13.2]
Qwen2-VL-7B	HR-4K	28	+4.5 [+1.8, +7.2]	+8.0 [+4.0, +12.2]	+1.0 [−2.8, +4.8]
Qwen2-VL-7B	CV-Bench	28	−3.4 [−5.0, −1.9]	—	—
Qwen2-VL-7B	RealworldQA	34	+0.0 [−6.2, +6.2]	—	—

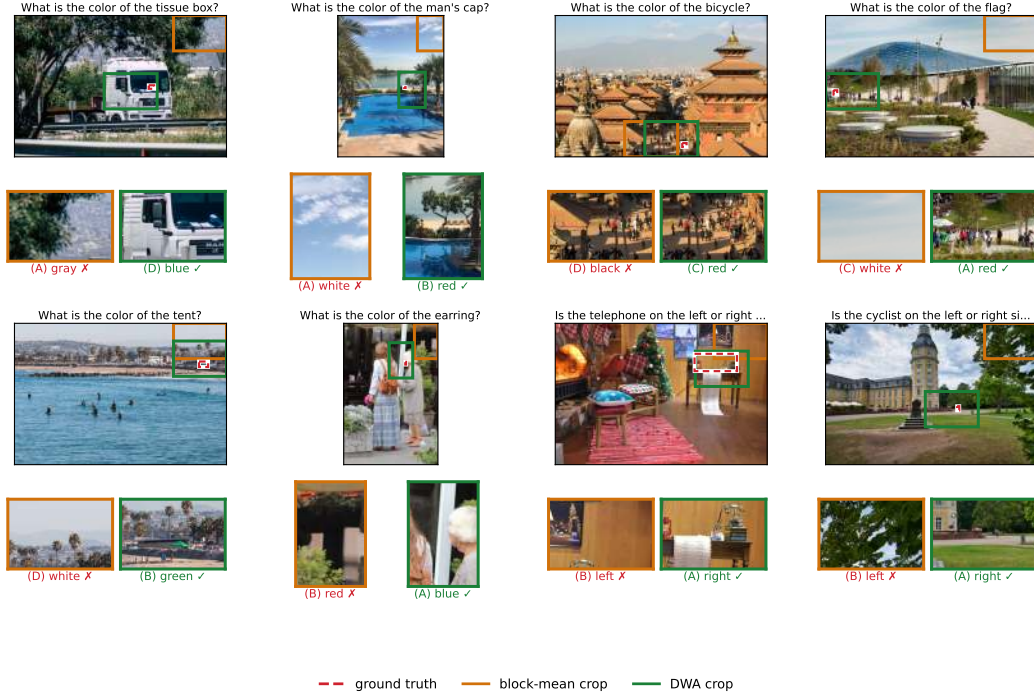


Figure 23: **The same eight-item view on Qwen2-VL-7B.** The failure mode is the same on a different model family and a different parameter count.

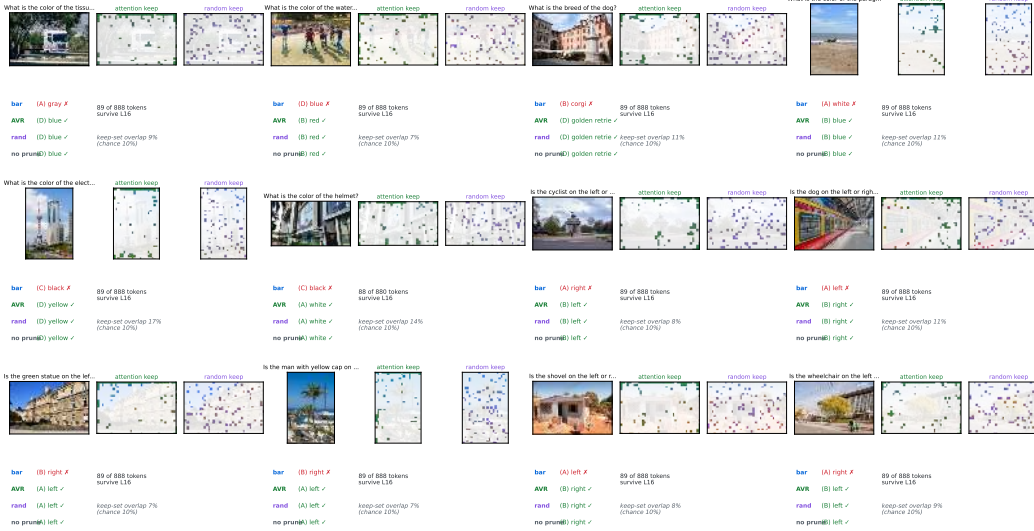


Figure 24: **Twelve inferences under AVR.** For each item: the image, the tokens kept by attention at the boundary, and an equally sized random keep. The two keep-sets overlap at close to the 10% chance rate and the model answers identically under both, and identically again without pruning at all.

T DESIGNS THAT DID NOT WORK

The last row generalises the rest, and explains why the two policies cannot be merged into one. DWA's signal is produced after the transport boundary, and pruning only saves compute at or before it. A pipeline that prunes early enough to save has destroyed the layers DWA reads, which costs 3.1

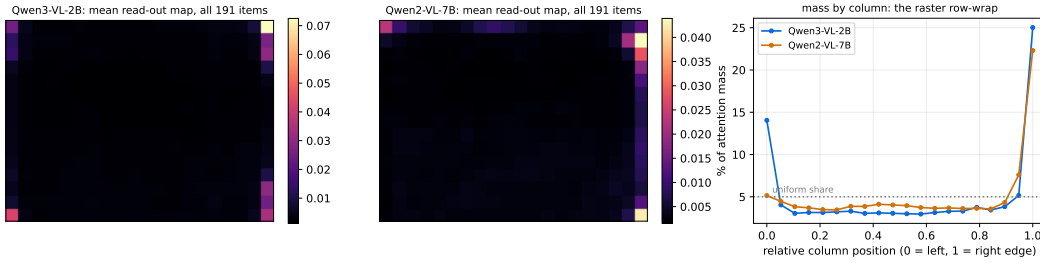


Figure 25: **The serialisation sink, averaged over the benchmark.** Each item’s read-out map is resampled to a common grid and averaged over all 191 items. Mass concentrates in the last column — the token before a raster row-wrap — on both checkpoints, at roughly four times the uniform share. This is what the outer-ring mask in every deployed implementation is patching around.

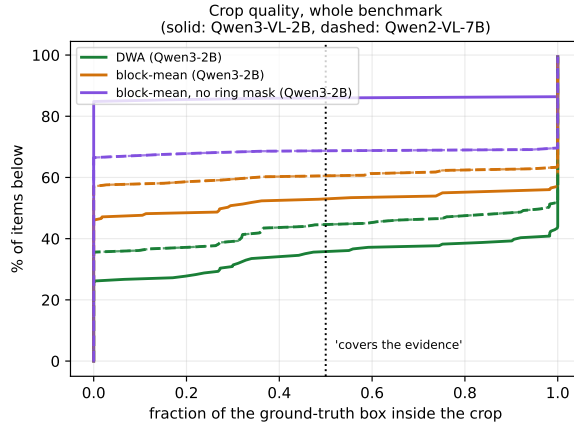


Figure 26: **Crop quality over the whole benchmark.** Cumulative distribution of the fraction of the ground-truth box captured by the crop. Without the ring mask the block-mean rule is close to useless; with it, it is usable; DWA is better than both and barely uses the mask.

and 12.6 points of coverage; a pipeline that waits for DWA has nothing left worth pruning. Holding the ranking fixed and moving only the cut from L16 to L4 costs 5.2 points. The incompatibility is structural, not a tuning failure.

The two policies are substitutes, not complements. The structural argument above is not the only reason they do not combine. We also tested the composition that *is* mechanically sound: hold the token-layer budget fixed, and let any placement rule spend its answer pass on a crop encoded at 460 tokens and pruned 90% at the boundary instead of a plain crop at 300. Applied to ViCrop-style block-mean, to LASER, to DWA and to oracle placement, on both checkpoints, every one of the eight contrasts spans zero. The diagnostic says why: the resolution headroom left *on a crop* averages +0.8 points, against +4.5 on the full image. A 0.25×0.25 window re-encoded at 300 tokens has already lifted the object above the encoding cliff, so the magnification AVR would buy has already been spent. Both policies spend the same resource, effective resolution on the region that matters; cropping spends it in a targeted way and AVR spends it uniformly. That is why their *scope* is complementary while their *gains* are not additive.

Why cropping cannot reach cross-instance questions. Under *perfect* placement, cross-instance accuracy reaches 75.0 and 77.6, an irreducible error of 25% and 22%, against 3.5% and 6.1% for single-instance. Two crops at full resolution covering *both* objects, which raises cross-instance object coverage by 17.1 and 15.8 points, still score -2.6 and -3.9 against a single crop. Cross-instance error is not predominantly perceptual, so no reallocation of perceptual resources reaches it. Resolution does reach part of it, which is what the second policy exploits, but cropping does not.

Table 38: The ten designs intended to extend cropping to cross-instance questions, and the two that tried to combine the two policies. Each was pre-registered with a primary contrast and a guard before it ran. “Cleared” counts checkpoints of two on which the primary passed.

design	what it changed	cleared	diagnosed cause
per-item window sizer	crop size from map features	0/2	grid inversion, weak features
peak-spanning window	box spanning the top peaks	0/2	relative signal, absolute constant
mass-containment window	window holding $x\%$ of mass	0/2	boxes far too large
joint cell-window ranking	arg-max over both at once	0/2	metric artefact: coverage rises with size
expected-accuracy window	window by predicted accuracy	0/2	accuracy-size curve flat against noise
thumbnail in answer pass	crop plus 64-token scene	0/2	second object below the encoding cliff
global-local split	scene@150 plus crop@150	0/2	halving the crop costs more than the scene gives
context-continued crop	scene@300 plus crop@300	0/2	scene costs 8.7 and 5.2 even with a perfect crop
multi-crop, matched budget	2×150 and 3×100	0/2	magnification tax (-11.3 , -4.3) exceeds coverage gain ($+7.8$, $+2.6$)
funded multi-crop	2×300 , pruned, early exit	0/2	$+3.7$ on one model, -2.6 on the other
DWA-guided early pruning	rank at L4 instead of L16	0/2	-5.2 : ranking before transport is worthless
AVR stacked on a placement rule	crop@460 pruned at L16, same budget	0/2	no resolution headroom left on a crop ($+0.8$ vs $+4.5$)

U PIPELINE FAULTS FOUND, AND HOW THEY WERE CAUGHT

Two runs in this work produced plausible numbers that were wrong. We report them because the rules that caught them are the reason we trust the rest, and because both faults are easy to make and invisible in aggregate accuracy.

A silently unpatched module. Pruning is injected by monkey-patching the attention forward pass. The patch covered two module paths; a third checkpoint’s attention lived in a third module, so pruning was a no-op and the arm was byte-identical to its unpruned control while reporting a clean $+12.2$. It was caught by a standing rule: a contrast of exactly $+0.0$ with a $[+0.0, +0.0]$ interval is a pipeline fault, never a result, because real effects do not land on zero to machine precision. The fix asserts at load time that the model’s actual attention module is one the patch reached, and prints it every run.

A hard-coded depth on a differently-shaped model. The arm table hard-coded the prune layer at 16. On the 36-layer checkpoint the correct depth, $0.57 \times 36 = 21$, was computed and printed but never used, so the run pruned at 44% of depth, inside the transport window. It was caught by the compute budget reading 78% of the bar instead of the expected 98%. We retain the voided run as accidental evidence for the boundary itself: pruning inside the window cost -7.9 pooled and -10.5 cross-instance on a third model family.

Every arm therefore prints its realised token count and its token-layer budget as a fraction of the bar, and any run outside 96–104% is discarded rather than reported.